

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



Le Xuan An

**Gradient-Routed Alignment with Capacity-aware Experts
for Multimodal Understanding**

KHOÁ LUẬN TỐT NGHIỆP

Ngành: Khoa học máy tính

Giảng viên hướng dẫn: Tiến sĩ Nguyễn Văn Sơn

Giảng viên đồng hướng dẫn: Phó Giáo sư Võ Đình Hiếu

HÀ NỘI - 2026

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



Le Xuan An

Gradient-Routed Alignment with Capacity-aware Experts
for Multimodal Understanding

KHOÁ LUẬN TỐT NGHIỆP

Ngành: Khoa học máy tính

Giảng viên hướng dẫn: Tiến sĩ Nguyễn Văn Sơn

Giảng viên đồng hướng dẫn: Phó Giáo sư Võ Đình Hiếu

HÀ NỘI - 2026

VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY



Le Xuan An

**Gradient-Routed Alignment with Capacity-aware Experts
for Multimodal Understanding**

BACHELOR'S THESIS

Major: Computer Science

Advisor: Dr. Nguyen Van Son

Co-Supervisor: Associate Professor Vo Dinh Hieu

HANOI - 2026

Lời cảm ơn

Lời đầu tiên, tôi xin gửi lời cảm ơn chân thành và lòng biết ơn sâu sắc tới thầy TS. Nguyễn Văn Sơn và thầy PGS.TS. Võ Đình Hiếu vì đã hướng dẫn và hỗ trợ tôi về kiến thức chuyên môn vô cùng tận tình trong quá trình nghiên cứu và thực hiện khóa luận này.

Tôi cũng xin được gửi lời cảm ơn đến thầy cô, các anh chị và các bạn trong phòng thí nghiệm *iSE - Intelligent Software Engineering* đã luôn hỗ trợ tôi về kiến thức và chuyên môn trong quá trình làm khoá luận.

Cuối cùng, tôi xin gửi lời cảm ơn chân thành tới các thầy, cô trong trường Đại học Công Nghệ - Đại học Quốc gia Hà Nội đã tạo điều kiện cho tôi được học tập và nghiên cứu để có thể hoàn thành khoá luận với kết quả tốt nhất.

Assurance

I hereby affirm that this thesis is my own original work. I have not submitted it previously for any academic or professional award.

I completed this thesis through my own efforts, with guidance and support from Dr. Nguyen Van Son and Associate Professor Vo Dinh Hieu, in partial fulfillment of the graduation requirements. I confirm that all information, data and findings presented here are original and free from plagiarism.

I have acknowledged every source that I consulted and included each one in the references. I accept full responsibility for the accuracy and integrity of this thesis under the regulations of VNU University of Engineering and Technology.

Hanoi, Ngày 20 tháng 5 năm 2026

Student

Le Xuan An

Tóm tắt

Khóa luận này nghiên cứu bài toán chưng cất tri thức đa giáo viên khác kiến trúc cho mô hình thị giác-ngôn ngữ (VLM). Mục tiêu là truyền tri thức từ một tập mô hình giáo viên đồng băng sang một mô hình học sinh nhỏ hơn khi các giáo viên thuộc những họ mô hình khác nhau, sử dụng bộ tách từ khác nhau và có mức độ hữu ích thay đổi theo từng mẫu. Trong bối cảnh này, chưng cất trực tiếp theo chỉ số token không còn hợp lệ và phép lấy trung bình đều giữa các giáo viên không giải quyết được khó khăn thứ ba: nó trộn lẫn tín hiệu hữu ích với tín hiệu xung đột mà không phân biệt.

Để giải quyết bài toán trên, khóa luận đề xuất Gradient-Routed Alignment with Capacity-aware Experts (GRACE), một khung chưng cất tri thức VLM đa giáo viên khác kiến trúc. GRACE kết hợp ba cơ chế: chưng cất Byte-Trie Wasserstein trên cây byte UTF-8 dùng chung, định tuyến giáo viên thừa top- k dựa trên trạng thái ẩn của mô hình học sinh và điều chỉnh trọng số theo độ đồng thuận gradient. Hàm mất mát Byte-Trie Wasserstein so sánh phân phối đầu ra của giáo viên và học sinh giữa các bộ tách từ khác nhau. Định tuyến thừa chọn các giáo viên phù hợp với từng mẫu. Cơ chế đồng thuận gradient sau đó điều chỉnh trọng số đã định tuyến theo mức độ căn chỉnh giữa gradient chưng cất của từng giáo viên và gradient của tổn thất CE có giám sát.

Thực nghiệm trên TextVQA, DocVQA và ChartQA đánh giá GRACE với hai mô hình học sinh SmolVLM-500M và SmolVLM-256M với tập bốn giáo viên khác kiến trúc. Với SmolVLM-500M, GRACE đạt 69.38, 73.11 và 80.20 trên TextVQA, DocVQA và ChartQA, thu hẹp lần lượt 55.03%, 39.53% và 55.34% khoảng cách ban đầu tới giáo viên có điểm cao nhất. Với SmolVLM-256M, GRACE đạt 60.73, 60.48 và 72.32 trên cùng ba bộ đánh giá. Nhìn chung, GRACE vượt qua tinh chỉnh trực tiếp, chưng cất logits cùng họ mô hình, ULD khác họ mô hình và các phương pháp đa giáo viên không đồng nhất được so sánh.

Từ khóa: mô hình thị giác-ngôn ngữ, chưng cất tri thức, chưng cất đa giáo viên, định tuyến giáo viên thừa, đồng thuận gradient, chưng cất Wasserstein khác bộ tách từ

Abstract

Distilling a compact VLM from a heterogeneous teacher pool faces three difficulties simultaneously: mismatched tokenizers make direct KD invalid, teacher usefulness varies sample by sample and a routed teacher can still produce gradients that oppose the supervised objective. Fixed teacher averaging fails to address this third difficulty: it mixes helpful signals with conflicting ones without discrimination.

Gradient-Routed Alignment with Capacity-aware Experts (GRACE) addresses all three. It combines Byte-Trie Wasserstein knowledge distillation over a shared UTF-8 byte trie, sparse top- k teacher routing conditioned on student hidden states and gradient-agreement refinement of routed teacher weights. The Byte-Trie Wasserstein loss compares teacher and student output distributions across tokenizers without shared-index assumptions, while sparse routing and gradient-agreement refinement select teachers per sample and down-weight those whose KD gradients conflict with the CE objective.

Experiments on TextVQA, DocVQA and ChartQA evaluate GRACE with SmolVLM-500M and SmolVLM-256M students with a four-teacher heterogeneous pool. For SmolVLM-500M, GRACE reaches 69.38, 73.11 and 80.20 on TextVQA, DocVQA and ChartQA, closing 55.03%, 39.53% and 55.34% of the original gap to the highest-scoring teacher. For SmolVLM-256M, GRACE reaches 60.73, 60.48 and 72.32 on the same benchmarks. In all comparisons, GRACE improves over direct fine-tuning, same-family logit-based KD, cross-family ULD and the heterogeneous multi-teacher baselines.

Keywords: vision-language model, knowledge distillation, multi-teacher distillation, sparse teacher routing, gradient agreement, cross-tokenizer Wasserstein distillation

Table of Contents

Lời cảm ơn

Assurance i

Tóm tắt ii

Abstract iii

Table of Contents iv

List of Figures vi

List of Tables vii

List of Abbreviations viii

Chapter 1 Introduction 1

Chapter 2 Background and Related Work 5

2.1 Vision-Language Models 5

2.2 Knowledge Distillation 7

2.3 Tokenizer-Agnostic Distribution Matching 8

2.4 Sparse Routing and Expert Selection 11

2.5 Multi-Teacher Distillation 13

2.6 Related Work 14

Chapter 3 GRACE: A Heterogeneous Multi-Teacher Distillation Framework 15

3.1 Method Overview and Problem Formulation 15

3.2 Byte-Trie Wasserstein Distillation 17

3.2.1 Shared Byte Trie 18

3.2.2 Sparse Support and Residual Tail Edge 19

3.2.3 Byte-Trie Wasserstein Objective 21

3.3	Teacher Routing Network	22
3.4	Gradient-Agreement Refinement	25
3.4.1	Agreement Score Computation	25
3.4.2	Warmup, EMA Smoothing and Active Set	26
3.4.3	Fallback, Uniform and Blended Weighting	27
3.5	Routed KD Aggregation and Final Objective	29
Chapter 4 Evaluation Methodology		32
4.1	Experimental Setup	32
4.2	Datasets	33
4.3	Baselines	34
4.4	Ablation Settings	37
Chapter 5 Experiment Results		38
5.1	Main Comparison Results	38
5.2	Ablation Study	43
5.3	Efficiency Analysis	44
5.4	Discussion	45
Conclusion and Future Work		50
References		51
Appendix A Proof of the Closed-Form Trie-Wasserstein		57
Appendix B Implementation Details		60
B.1	Baseline Implementation	61
B.2	Offline Teacher-Logit Preparation	62
B.3	Remote Logit Streaming During Training	63

List of Figures

2.1	Representative architecture families of modern vision-language models. . . .	6
2.2	Geometric behavior of Wasserstein distance.	9
2.3	Discrete optimal transport views.	10
3.1	Overview of the GRACE architecture.	16
3.2	Shared byte-trie tokenizer matching.	19
3.3	Sparse Byte-Trie Wasserstein support.	21
3.4	Teacher routing network.	23
3.5	Top- k routing dynamics.	24
3.6	EMA smoothing.	26
3.7	GRACE score refinement.	27
3.8	Representative GRACE weighting cases.	28
3.9	Sample-wise weight refinement.	29
3.10	Sample-specific weighting.	29
4.1	Representative dataset examples.	33
5.1	Grouped benchmark comparison.	40
5.2	Uniform and GRACE training curves.	41
5.3	Routed-teacher load examples.	41
5.4	Final-layer CKA heatmaps.	47
5.5	Vision-layer CKA heatmaps.	47
5.6	Representative TextVQA attention example.	49
B.1	R2 teacher-logit storage.	63
B.2	Training with remote teacher logits.	64

List of Tables

2.1	Summary of response-level and feature-level KD objectives.	8
3.1	Notation table.	16
4.1	Dataset split sizes.	34
4.2	Baseline groups.	35
4.3	Student and teacher model results.	36
5.1	Main benchmark comparison.	39
5.2	Relative gains of GRACE.	42
5.3	Gap-to-teacher analysis for SmolVLM-500M.	42
5.4	DocVQA component ablation.	43
5.5	ULD vs. Byte-Trie Wasserstein under equal ranked masses.	46
B.1	Experiment parameters.	60
B.2	Logit-streaming settings.	64

List of Abbreviations

Abbr.	Full Term	Description
ANLS	Average Normalized Levenshtein Similarity	Evaluation metric used for DocVQA.
CE	Cross-Entropy	Supervised answer-token loss used during training.
CKA	Centered Kernel Alignment	Representation similarity measure used to compare hidden layers.
ChartQA	Chart Question Answering	Benchmark for question answering over charts.
DocVQA	Document Visual Question Answering	Benchmark for question answering on document images.
EOS	End of Sequence	Special token that marks the end of a generated answer span.
GRACE	Gradient-Routed Alignment with Capacity-aware Experts	Proposed multi-teacher distillation method in this thesis.
SHM-CKA	Hidden-State Matching with CKA	Baseline that combines logits distillation with hidden-state CKA matching.
JS	Jensen-Shannon	Symmetric divergence measure between probability distributions.
KD	Knowledge Distillation	Transfer of supervision from teacher models to a student model.
KL	Kullback-Leibler	Divergence used to compare probability distributions.
LLM	Large Language Model	Language model with large parameter scale and generation capability.
MLP	Multi-Layer Perceptron	Feedforward projection block used in the teacher routing network.
MoE	Mixture of Experts	Sparse routing framework that motivates teacher selection.

Abbr.	Full Term	Description
MTKD	Multi-Teacher Knowledge Distillation	Distillation setting that uses more than one teacher model.
OCR	Optical Character Recognition	Extraction of text from images or scanned documents.
OT	Optimal Transport	Distribution-matching view used to motivate trie-Wasserstein KD.
TWD	Tree-Wasserstein Distance	Tree-structured optimal-transport distance used as background for the proposed loss.
TextVQA	Text Visual Question Answering	Benchmark for question answering on natural images containing text.
ULD	Universal Logit Distillation	Tokenizer-agnostic distillation baseline based on ranked probability distributions.
VLM	Vision-Language Model	Model that processes visual and textual inputs together.
VLM-Eval-Kit	Vision-Language Model Evaluation Kit	Toolkit used to run standardized multimodal evaluation.
VQA	Visual Question Answering	Task of answering questions grounded in visual inputs.

Chapter 1

Introduction

The ability to reason jointly over images and text has become central to a wide range of real-world tasks. Automated document processing pipelines extract tables, figures and key fields from millions of scanned pages daily. Financial and scientific workflows depend on systems that can read charts and answer quantitative questions about them. Scene-text understanding is increasingly embedded in consumer devices, enabling features from translation overlays to accessibility tools. Getting a model to read text in an image or answer a question about a chart was a research challenge just a few years ago, but it is now a core component of production systems. Across all of these settings, the underlying requirement is the same: a model that can interpret visual content and produce accurate, grounded text responses. **Vision-language models (VLMs)** have emerged as the leading approach, combining a vision encoder, a multimodal connector and an autoregressive language decoder into a unified architecture capable of handling this full range of tasks [1–6]. Over the past few years these models have advanced rapidly: modern systems can parse receipts, describe medical images and answer free-form questions about charts with accuracy that matches or surpasses specialist tools on several benchmarks [7–9].

This progress, however, has come with a steep increase in model size. Neural scaling laws favour larger language backbones and the most capable open-weight VLMs, such as Qwen2.5-VL-72B and Llama 3-V 405B, require tens to hundreds of gigabytes just to store their weights, well beyond the memory budget of a single GPU [7, 10–12]. Serving them demands multi-GPU clusters that are incompatible with the latency, cost and privacy constraints of deployment in edge devices, mobile applications, robotics and on-premise enterprise systems [13, 14]. The result is a growing gap between what the best models can do and what can realistically be deployed: a model that can read a receipt or interpret a chart is only useful if it can actually run where those tasks arise and for most practical settings the frontier models cannot.

Knowledge distillation offers a principled route to closing this gap. Rather than training a compact model from scratch on *hard labels*, the student is trained to match the teacher’s *soft output distributions*, which carry probability signals over alternative answers that provide far richer supervision than one-hot targets [15, 16]. Recent lightweight VLMs

produced by distillation reach competitive benchmark scores at a fraction of the teacher’s compute cost [17–19]. A natural and attractive extension is to distill from a *pool* of teachers drawn from different model families, each specialising in different visual domains. In practice, no single model is the strongest teacher across all task types: one family may have been pretrained on richer document corpora, making it a better supervisor for layout-heavy samples, while another may have stronger chart or scene-text representations. Pooling their supervision could expose a single compact student to a breadth of complementary capabilities that no individual teacher can provide on its own.

In practice, however, heterogeneous multi-teacher distillation is harder than it first appears and the challenges compound in ways that existing methods do not anticipate. Teachers from different families use incompatible tokenizers, making the standard approach of comparing output distributions at shared vocabulary indices invalid. Teacher usefulness also varies per sample: the strongest supervisor for a chart question is rarely the best choice for a scanned receipt or a scene-text image, so fixed-weight averaging dilutes or wastes the most relevant signals. Most critically, even a well-matched teacher can produce KD gradients that directly oppose the supervised learning objective, degrading the student instead of improving it. These three constraints, **tokenizer mismatch**, **per-sample teacher relevance** and **gradient conflict**, must be resolved simultaneously, yet no existing method addresses all three at once [20–22].

Gradient-Routed Alignment with Capacity-aware Experts (GRACE) treats each of the three constraints as a first-class design requirement, not an edge case. To handle tokenizer mismatch, it encodes all decoded tokens as UTF-8 byte sequences and compares teacher and student output distributions through a shared byte trie. Tokens that spell similar strings incur low transport cost regardless of which vocabulary produced them, giving a meaningful signal even when teacher and student tokenizers are completely different. To handle per-sample teacher relevance, a lightweight router conditioned on the student’s own hidden states selects a small subset of teachers for each input, so the supervision mix adapts to the content of the image and question instead of remaining fixed across all training samples. To handle gradient conflict, a refinement step measures how well each selected teacher’s KD gradient aligns with the supervised CE gradient and down-weights any teacher whose signal would pull the student away from the correct answer even when that teacher appears relevant.

Together, these three components form a distillation framework in which teacher diversity is genuinely put to use: each sample receives supervision from the teachers most suited to it, filtered by both content relevance and optimization compatibility. With a 500M-parameter

student and a four-teacher pool drawn from three model families, GRACE closes more than half the original teacher-student gap on two of three benchmarks. This result suggests that capable VLMs can be built on a single GPU and that the primary obstacle to doing so is not model size but the quality and consistency of the distillation signal. This thesis investigates three questions to characterise where these gains come from:

- **RQ1: Performance.** How does GRACE compare with direct fine-tuning, same-family logit-based KD, cross-tokenizer single-teacher distillation and heterogeneous multi-teacher baselines on the target VQA benchmarks?
- **RQ2: Component Analysis.** How much do the cross-tokenizer distillation loss, the routing mechanism and the gradient-conflict resolution step contribute to the final performance?
- **RQ3: Efficiency.** What runtime overhead does GRACE introduce during training?

This thesis makes three main contributions:

- **Byte-Trie Wasserstein distillation.** We propose a KD loss that maps teacher and student tokens onto a shared UTF-8 byte trie and compares output distributions through byte-path proximity. Without this, direct output comparison across heterogeneous teachers is invalid. Applying standard KL divergence when teacher and student have different vocabularies silently compares unrelated token indices. The proposed loss removes this assumption entirely, enabling distillation from any teacher regardless of its tokenizer or vocabulary size.
- **Objective-aware multi-teacher weighting.** Top- k routing selects candidate teachers per sample, gradient agreement then filters their KD signals against the supervised CE objective, suppressing teachers whose gradients conflict with learning.
- **Empirical validation on VQA benchmarks.** We evaluate GRACE on **TextVQA**, **DocVQA** and **ChartQA** with two SmolVLM students and a four-teacher heterogeneous pool. GRACE outperforms all compared baselines, with SmolVLM-500M it closes 55%, 40% and 55% of the original teacher-student gap on the three benchmarks.

The thesis is organized as follows. Chapter 2 reviews Vision-Language Models (VLMs), Knowledge Distillation (KD), cross-tokenizer matching, sparse routing and gradient conflict resolution. Chapter 3 presents GRACE in detail. Chapter 4 describes the datasets, evaluation metrics, baselines and ablation settings. Chapter 5 reports the main comparative

results, ablation studies, efficiency analysis and discussion. Finally, Chapter 5 summarizes the key findings and discusses limitations and directions for future work.

Chapter 2

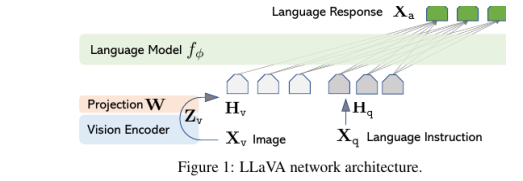
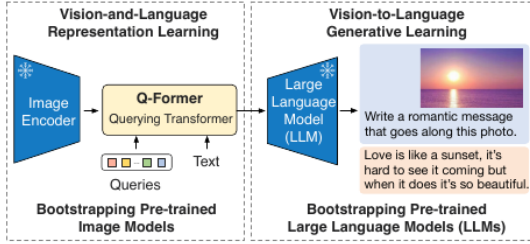
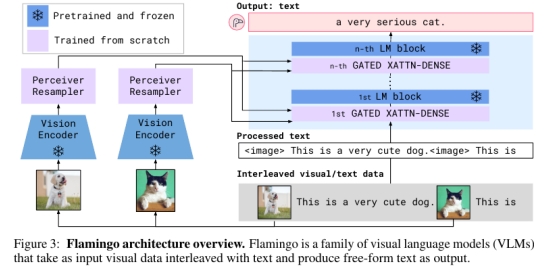
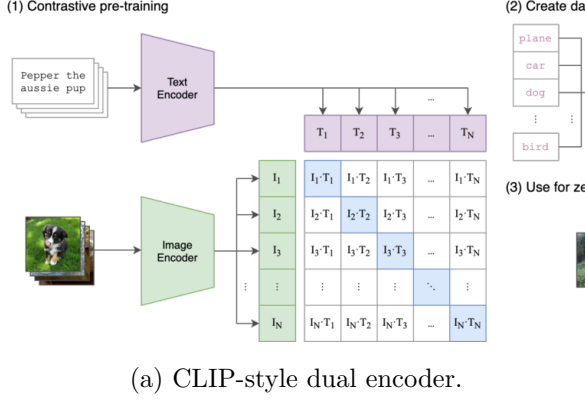
Background and Related Work

Three bodies of prior work bear directly on this thesis: knowledge distillation, tokenizer-agnostic distribution matching and sparse expert routing. The final section surveys where these lines of work converge and the gap that motivates GRACE.

2.1 Vision-Language Models

Modern vision-language models combine a vision encoder with a language model. On the visual side, many systems use a **Vision Transformer** pretrained with an image-text objective, commonly **CLIP** [23] or **SigLIP** [24], which converts an image into a sequence of patch embeddings. A connector module then fuses these visual tokens with text, using cross-attention, a **Q-Former** or a learned projector [1–3]. This design separates visual encoding from text generation: the vision encoder produces a compact sequence of image features, while the language model generates answers token by token. VLM families differ in where and how vision-language fusion happens, as shown in Figure 2.1. Dual-encoder models align image and text embeddings contrastively, cross-attention models let language tokens attend to visual features directly and projector-based models project visual embeddings into the language model’s input space and decode normally. This thesis focuses on projector-based generative VLMs because their autoregressive output distributions are what GRACE distills. Projected image embeddings are inserted as visual tokens, so answer generation is standard next-token prediction over combined visual and text context.

²Panels adapted from CLIP [23], Flamingo [1], BLIP-2 [2] and LLaVA [3].



(c) BLIP-2 with Q-Former.

(d) LLaVA projector-based design.

Figure 2.1: Representative architecture families of modern vision-language models.²

Most released projector-based VLMs are instruction-tuned on visual question-answering data after pretraining, making them capable of producing short, grounded answers before any distillation is applied [3, 19]. All student and teacher checkpoints used in this thesis are instruction-tuned variants of their respective base models.

The student VLM produces answer tokens autoregressively. The Transformer maps each hidden state $\mathbf{h}_j$ to next-token logits and probabilities [25]:

$$\mathbf{z}_j = \mathbf{W}_o \mathbf{h}_j + \mathbf{b}_o, \quad p_j(v) = \text{softmax}(\mathbf{z}_j)_v = \frac{\exp(z_{j,v})}{\sum_{u \in \mathcal{V}} \exp(z_{j,u})}. \quad (2.1)$$

For an answer sequence $y = (y_1, \dots, y_L)$ with multimodal input x , the model factorizes the conditional probability as

$$p_\theta(y \mid x) = \prod_{j=1}^L p_\theta(y_j \mid x, y_{<j}). \quad (2.2)$$

Token-level distillation compares teacher and student answer-token distributions, $p_{t,j}$ and $p_{s,j}$, at each supervised position [15, 25]. In VQA, short answers such as numbers, names and document fields can differ by a single token, so each position carries real weight.

2.2 Knowledge Distillation

Knowledge distillation trains a compact student to match the outputs of a larger teacher [15, 16]. Let $\mathcal{L}_{CE}$ denote the supervised cross-entropy loss of the student. The combined objective is

$$\mathcal{L} = \lambda_{CE}\mathcal{L}_{CE} + \lambda_{KD}\mathcal{L}_{KD}, \quad (2.3)$$

where $\mathcal{L}_{KD}$ is a teacher-guided loss. The student thereby learns the teacher’s relative probabilities over alternative answers, often called **dark knowledge**, which hard labels discard [15]. A common same-vocabulary KD loss is

$$\mathcal{L}_{KD} = \tau^2 \text{KL}\left(\mathbf{p}_t^{(\tau)} \parallel \mathbf{p}_s^{(\tau)}\right), \quad (2.4)$$

where τ is the *distillation temperature*. The cross-entropy term anchors the student to the ground-truth answer, while the KD term transfers the teacher’s relative preference over alternative tokens. In the heterogeneous setting, the KD term must be redefined to handle mismatched tokenizers.

Let $\mathbf{z}_s$ and $\mathbf{z}_t$ denote student and teacher logits. With temperatures τ_s and τ_t , the softened distributions are

$$\mathbf{p}_s = \text{softmax}\left(\frac{\mathbf{z}_s}{\tau_s}\right), \quad \mathbf{p}_t = \text{softmax}\left(\frac{\mathbf{z}_t}{\tau_t}\right). \quad (2.5)$$

Classical KD divergences, including forward KL and Jensen-Shannon, assume that teacher and student share the same vocabulary [15, 20, 26]. The temperature controls probability mass beyond the top answer token: higher τ flattens the distribution and spreads more mass to lower-ranked alternatives, while lower τ sharpens it toward hard-label supervision. The teacher’s tail often encodes semantic alternatives such as related words, number formats or subword variants.

For multimodal question answering, distillation is restricted to the supervised answer positions, the span evaluated at test time [3]. The KD loss therefore masks ignored tokens and truncates student and teacher answer spans to a common length. Let Ω_b^{CE} denote the non-ignored student answer positions for sample b , so the supervised loss is

$$\mathcal{L}_{CE}^{(b)} = \frac{1}{|\Omega_b^{CE}|} \sum_{j \in \Omega_b^{CE}} -\log p_s(y_j \mid x_b, y_{<j}). \quad (2.6)$$

The same masking applies to KD positions, keeping the teacher signal on the evaluated answer span and not on prompt reconstruction, padding or formatting. For multiple

teachers, the KD position set can be teacher-specific because teacher tokenizers and generated answer spans may differ. Per-teacher KD positions $\Omega_{b,i}^{KD}$ cover only the valid aligned span for each teacher.

Table 2.1 lists common response-level divergences and feature-level losses [15, 26]. In the table, p and q denote teacher and student output distributions over token events t and $m = \frac{1}{2}(p + q)$ is the JS midpoint distribution. The teacher and student feature vectors are

$$\mathbf{u} = \Phi_T(f_T(x, y)), \quad \mathbf{v} = \Phi_S(f_S(x, y)).$$

The index r marks feature dimensions or normalized feature bins. *Response-based* distillation trains on the teacher’s output distribution at supervised positions, the classical form of [15], still standard in language-model and VLM compression [16, 27].

Feature-based distillation matches intermediate hidden states. Given student and teacher hidden states $\mathbf{H}_s^{(\ell)}$ and $\mathbf{H}_t^{(m)}$ at selected layers, a matching objective is

$$\mathcal{L}_{feat} = \left\| \phi_s(\mathbf{H}_s^{(\ell)}) - \phi_t(\mathbf{H}_t^{(m)}) \right\|_2^2, \quad (2.7)$$

where ϕ_s and ϕ_t are projection functions that make the hidden states comparable. Feature-based methods sidestep the vocabulary problem: the comparison operates on hidden states, not token indices [28, 29]. But they do not supervise answer tokens directly and can transfer architecture-specific representations, so neither signal is uniformly better [28, 30].

Table 2.1: Summary of response-level and feature-level KD objectives.

Response-Level Divergences		Feature-Level Losses	
Type	$D(p, q)$	Function $\mathcal{L}_F$	Expression
Forward KLD	$\sum_t p(t) \log \frac{p(t)}{q(t)}$	L2-Norm Distance	$\ \mathbf{u} - \mathbf{v}\ _2$
Reverse KLD	$\sum_t q(t) \log \frac{q(t)}{p(t)}$	L1-Norm Distance	$\ \mathbf{u} - \mathbf{v}\ _1$
JS Divergence	$\frac{1}{2} \text{KL}(p\ m) + \frac{1}{2} \text{KL}(q\ m)$	Cross-Entropy Loss	$-\sum_r \mathbf{u}_r \log \mathbf{v}_r$

2.3 Tokenizer-Agnostic Distribution Matching

Same-vocabulary KD assumes that teacher and student probabilities are indexed by the same token set. Heterogeneous VLM distillation breaks this assumption because a teacher

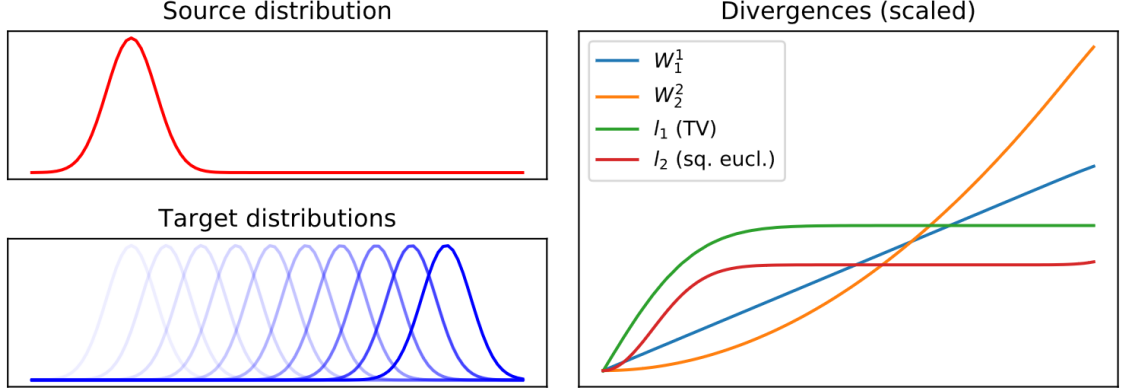


Figure 2.2: Geometric behavior of Wasserstein distance.¹

may tokenize the string “invoice” as one token while the student splits it into several subword pieces, so directly applying KL over their respective vocabularies then compares unrelated indices and is invalid [20]. Comparing probability mass over decoded strings, rather than vocabulary indices, avoids the index-alignment problem. Let $\mathcal{V}_t$ and $\mathcal{V}_s$ be the teacher and student vocabularies with decoding maps into a shared string space $\mathcal{S}$:

$$\psi_t : \mathcal{V}_t \rightarrow \mathcal{S}, \quad \psi_s : \mathcal{V}_s \rightarrow \mathcal{S}.$$

Transport-based objectives replace same-vocabulary KL with a distance over probability mass on a shared string support.

Optimal transport compares two probability measures by the minimum cost required to move mass from one to the other [31]. For probability measures μ_s and μ_t , the p -Wasserstein cost is

$$W_p^p(\mu_s, \mu_t) = \min_{\gamma \in \Pi(\mu_s, \mu_t)} \int_{\Omega_s \times \Omega_t} \|x - y\|^p d\gamma(x, y), \quad (2.8)$$

where $\Pi(\mu_s, \mu_t)$ is the set of joint couplings with marginals μ_s and μ_t . For discrete weights $\mathbf{a}$, $\mathbf{b}$ and cost matrix $\mathbf{C}$, this becomes

$$W(\mathbf{a}, \mathbf{b}) = \min_{\mathbf{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle \mathbf{P}, \mathbf{C} \rangle, \quad \mathcal{U}(\mathbf{a}, \mathbf{b}) = \{ \mathbf{P} \geq 0 \mid \mathbf{P}\mathbf{1} = \mathbf{a}, \mathbf{P}^\top \mathbf{1} = \mathbf{b} \}. \quad (2.9)$$

Unlike pointwise divergences, Wasserstein distances reflect displacement even when supports are disjoint, as Figure 2.2 illustrates.

Figure 2.2 makes this geometric property concrete. The left panels show a fixed source distribution (red bell curve) and a sequence of target distributions (blue bell curves) that slide progressively further away. The right panel plots four distance measures as

¹Adapted from Flamary and Courty [32].

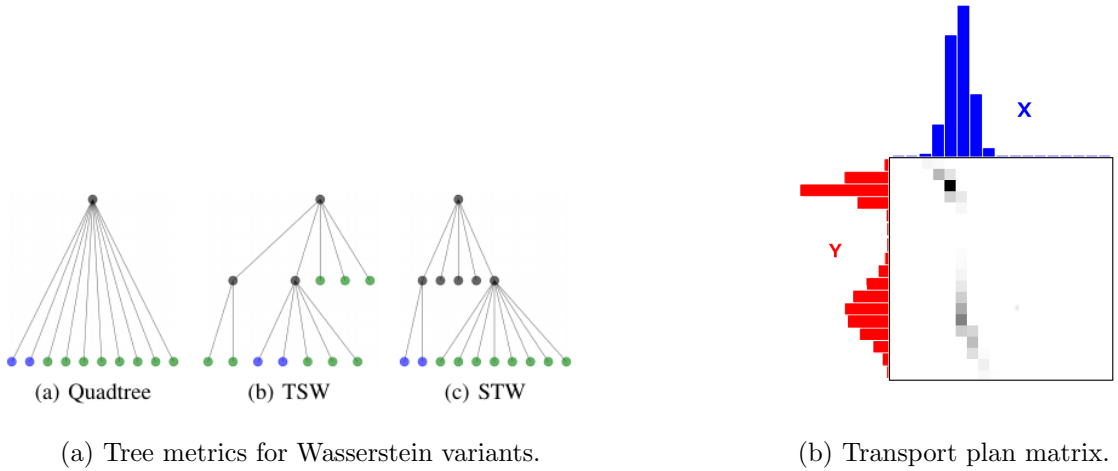


Figure 2.3: Discrete optimal transport views.¹

the shift increases. The ℓ_1 (total variation) and ℓ_2 (squared Euclidean) distances both **saturate** once the two distributions no longer overlap, providing no gradient signal about how far apart they are. The Wasserstein distances W_1 and W_2^2 , by contrast, grow monotonically with the displacement, reflecting the true cost of moving mass from one location to the other. This is precisely the property needed for cross-tokenizer distillation: teacher and student distributions rarely share support, so a loss that saturates when supports are disjoint carries no useful training signal.

For tokenizer-agnostic matching, dense OT is costly because it requires a pairwise ground cost between every source and target bin. A *tree metric* places tokens on a shared hierarchy instead and the distance reduces to weighted *subtree-mass imbalances* [31, 33]. For a rooted tree $T = (V, E)$ with edge weights $w(e)$, the distance is

$$W_T(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|, \quad (2.10)$$

where $\Gamma(e)$ denotes the subtree below edge e . On a tree metric, transport reduces to per-edge subtree-mass imbalances, avoiding the dense source-target coupling of standard OT [33, 34]. Figures 2.3a and 2.3b compare the tree-metric and matrix-coupling views of OT. Averaging over random tree metrics preserves OT geometry while admitting fast closed-form computation [34].

Representation-level alignment avoids the vocabulary problem by matching intermediate hidden states. Because sequence lengths and hidden dimensions differ across architectures, alignment requires pooling or projection before comparison. The resulting Frobenius-norm loss operates without any vocabulary assumption [28].

¹Panel (a) is adapted from Takezawa et al. [33]. Panel (b) is adapted from Wikimedia Commons and Peyré and Cuturi [31, 35].

Centered Kernel Alignment (CKA) compares representations through their similarity structure and avoids direct elementwise hidden-state matching [29]. Given activation matrices $\mathbf{X} \in \mathbb{R}^{n \times d_x}$ and $\mathbf{Y} \in \mathbb{R}^{n \times d_y}$ over the same n examples, the linear CKA score is

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{Y}^\top \mathbf{X}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \|\mathbf{Y}^\top \mathbf{Y}\|_F}. \quad (2.11)$$

Higher CKA values mean the two representations share the same pairwise similarity structure across the evaluated samples. Because CKA is invariant to isotropic scaling and orthogonal transformations, it remains valid when teacher and student have different hidden dimensions, unlike direct hidden-state matching. CKA serves here both as a distillation signal and as a diagnostic for feature alignment across heterogeneous architectures [29, 30].

2.4 Sparse Routing and Expert Selection

Recent sparse expert models route each sample or token through only a subset of available experts, activating only a fraction of parameters per input [36–38]. Let $\mathbf{h}^{(b)}$ denote a pooled representation for sample b . A router network $G_\phi(\cdot)$ produces gate logits

$$\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)}), \quad \tilde{\mathbf{r}}^{(b)} = \text{AdjustedScores}(\mathbf{r}^{(b)}), \quad \mathbf{p}^{(b)} = \text{softmax}(\tilde{\mathbf{r}}^{(b)}), \quad (2.12)$$

where $p_i^{(b)}$ is the initial routing weight for expert i . The adjusted scores $\tilde{\mathbf{r}}^{(b)}$ may include expert bias, training noise or temperature scaling before the routing softmax. **Top- k routing** retains only the highest-scoring experts and zeros out all others, while *capacity constraints* prevent any single expert from receiving too many assignments. The selected expert set is

$$\mathcal{S}_b = \text{TopK}(\mathbf{p}^{(b)}, k).$$

The output is a weighted sum over selected experts:

$$\mathbf{y}^{(b)} = \sum_{i \in \mathcal{S}_b} \alpha_i^{(b)} E_i(\mathbf{h}^{(b)}), \quad \alpha_i^{(b)} = \frac{p_i^{(b)}}{\sum_{j \in \mathcal{S}_b} p_j^{(b)}}. \quad (2.13)$$

Only the selected experts are evaluated, keeping active computation small even when the total expert count is large. Frozen teachers fit directly into this framework: each one contributes a weighted distillation loss for the samples routed to it.

Without regularization, sparse routing collapses onto a small subset of experts. In a sparse **Mixture of Experts** (MoE) layer, overloaded experts become the latency bottleneck while under-used ones receive few gradient updates and fail to specialize [36, 39, 40]. Let a mini-batch contain T routed items and let N denote the number of experts. For item b , the router

produces probabilities $\mathbf{p}^{(b)} \in \Delta^{N-1}$. A top- k router induces the binary assignment mask

$$\mathbf{a}_i^{(b)} = \mathbf{1}\{i \in \text{TopK}(\mathbf{p}^{(b)}, k)\}, \quad n_i = \sum_{b=1}^T \mathbf{a}_i^{(b)}, \quad (2.14)$$

where n_i is the number of routed items assigned to expert i . Balanced routing corresponds to

$$n_i \approx \frac{kT}{N}.$$

When n_i is much larger than this value for a few experts, the router has collapsed to a small ensemble of busy experts, leaving the rest idle. Auxiliary load-balancing losses track two quantities, *soft* load and *hard* load:

$$\bar{p}_i = \frac{1}{T} \sum_{b=1}^T p_i^{(b)}, \quad \bar{q}_i = \frac{1}{kT} \sum_{b=1}^T \mathbf{a}_i^{(b)}. \quad (2.15)$$

$\bar{p}_i$ is the average routing weight before hard selection, while $\bar{q}_i$ is the fraction of items the expert actually receives after top- k filtering. The **Switch Transformer** balancing loss uses their scaled dot product [37]:

$$\mathcal{L}_{bal} = \alpha_{bal} N \sum_{i=1}^N \bar{p}_i \bar{q}_i. \quad (2.16)$$

The factor N normalises for the number of experts. Since $\bar{q}_i$ is non-differentiable through the top- k decision, the loss penalizes concentrating probability on already-loaded experts through the differentiable $\bar{p}_i$ term. Entropy and z -loss terms stabilize sparse selection without dominating the task and distillation objectives.

Sparse routing is most effective when expert usefulness varies by input [36, 37]. In a generic *sample-adaptive* router, the selected expert set is a function of a sample representation:

$$\mathcal{S}_b = \text{TopK}(\text{softmax}(G_\phi(\mathbf{h}^{(b)})), k). \quad (2.17)$$

The loss is then computed only from experts in $\mathcal{S}_b$. This differs from *fixed ensembling*, where every expert contributes to every example with the same global weight. In multimodal QA, one sample may require OCR and document layout reasoning, another chart interpretation and a third natural image recognition.

In distillation, frozen teachers replace MoE experts: the router selects which ones supervise each sample and the routed distillation loss sums only over the selected set:

$$\mathcal{L}_{route}^{(b)} = \sum_{i \in \mathcal{S}_b} \omega_i^{(b)} \mathcal{L}_{KD}^{(b,i)}. \quad (2.18)$$

Since teachers are frozen, only the student and router parameters receive gradient updates. Routing by relevance does not guarantee gradient alignment: a highly-scored teacher can still push against the CE objective, so gradient-aware reweighting after routing is needed to filter misaligned signals [21, 41, 42].

2.5 Multi-Teacher Distillation

Let $\mathcal{T} = \{T_1, \dots, T_N\}$ be the teacher pool and $\mathcal{L}_{KD}^{(b,i)}$ the distillation loss from teacher T_i on sample b . The simplest aggregation rule is uniform averaging:

$$\mathcal{L}_{KD}^{(b)} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_{KD}^{(b,i)}. \quad (2.19)$$

Uniform averaging isolates teacher diversity from any learned selection, but equal influence is the wrong prior when teacher strengths vary across document images, charts and natural scene text.

Heterogeneous pools differ across multiple dimensions: tokenizer, language backbone, vision encoder, pretraining data and answer calibration. One model may provide stronger supervision for OCR-heavy samples while another is better calibrated on charts. But teacher distributions can also disagree on the same answer position when their tokenizers split the answer string differently. A multi-teacher method must therefore **select and weight teachers, not just aggregate them**.

Weighted aggregation generalizes this:

$$\mathcal{L}_{KD}^{(b)} = \sum_{i=1}^N w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}, \quad w_i^{(b)} \geq 0, \quad \sum_{i=1}^N w_i^{(b)} = 1. \quad (2.20)$$

Fixed weights ignore input variation entirely, while sample-dependent weights allow the best teacher for each input type to dominate. Section 2.6 covers prior methods in detail.

Sparse routing implements sample-dependent weighting directly: frozen teachers act as experts, each contributing a loss only for the samples routed to it. Routing introduces two problems that ensembling avoids: early collapse onto one teacher leaves the remaining teachers unused and a highly-ranked teacher can still push the gradient against the supervised CE objective.

This second issue is an **optimization conflict**, not just a selection problem. In the standard combined loss

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD},$$

both terms update the same parameters: the combined step is reinforced when CE and KD gradients agree and partially cancelled when they diverge [41, 42]. In multi-teacher distillation, each teacher contributes a separate KD gradient, so selecting a teacher subset does not guarantee the selected gradients align with the supervised update. GRACE uses a **two-stage design**: routing proposes candidate teachers, then a **post-routing compatibility check** verifies that their KD gradients align with the supervised CE objective.

2.6 Related Work

Four lines of prior work bear directly on GRACE, each addressing part of the problem without handling all three constraints together.

Response-based and vocabulary-mismatched KD. Standard response-based KD supervises the student with the teacher’s soft output distributions [15, 16, 27], but KL, reverse KL and Jensen-Shannon divergences all assume a shared vocabulary. Cross-tokenizer methods relax this by comparing probability mass over decoded strings: **ULD** [20] computes the L1 distance between rank-sorted probability vectors and tree-Wasserstein methods define a ground geometry over tokens without requiring index equality [33, 34], but both assume a single fixed teacher and provide no per-sample selection mechanism.

Representation matching and CKA alignment. Matching hidden states or CKA similarity structure sidesteps the vocabulary problem entirely and works across architectures with different hidden dimensions [28, 29]. However, similar representational structure does not imply similar answer distributions and these methods leave teacher selection unresolved. GRACE uses CKA as a diagnostic rather than a training signal, since the primary distillation objective is output-level.

Multi-teacher KD, routing and gradient-conflict resolution. Reinforcement-based and meta-learning methods learn sample-dependent teacher weights or selection policies [21, 22, 43], and sparse routing from mixture-of-experts models selects a teacher subset per input [36, 37], yet neither line addresses tokenizer mismatch. Gradient-direction filters [41, 42] resolve optimization conflict between auxiliary and primary losses, but prior multi-teacher methods do not combine them with cross-tokenizer matching or sparse selection.

The gap is structural: vocabulary-aligned KD breaks across tokenizers, cross-tokenizer transport assumes a single teacher, representation matching leaves teacher selection open, routing-based methods do not verify gradient compatibility. **GRACE closes all four gaps simultaneously.**

Chapter 3

GRACE: A Heterogeneous Multi-Teacher Distillation Framework

GRACE addresses three constraints that existing methods handle separately: **vocabulary mismatch** between teacher and student, **variable teacher usefulness** across samples and **KD gradients that conflict** with the supervised CE objective. The three components are introduced in this order, with each one addressing a constraint that the previous component leaves open.

3.1 Method Overview and Problem Formulation

We consider a training set

$$\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N_{\text{data}}},$$

where x_n is a multimodal input and y_n is the target answer sequence.

Let S_θ denote the student VLM with trainable parameters θ , let G_ϕ denote the trainable teacher router with parameters ϕ and let $\mathcal{T} = \{T_i\}_{i=1}^N$ be the frozen teacher pool. For each sample b , the student produces logits $\mathbf{z}_s^{(b)}$ and teacher T_i provides logits $\mathbf{z}_t^{(b,i)}$.

Teacher and student output spaces use different token indices, so naive KD compares unrelated vocabulary positions. Teacher usefulness is also sample-dependent: a model strong at document layout need not be strong at chart reasoning. Relevance alone is not enough: a teacher that appears well-matched to the sample can still produce a KD gradient that pushes against the supervised CE objective. GRACE addresses these constraints with **Byte-Trie Wasserstein** matching, **sparse teacher routing** and **gradient-agreement refinement**.

The full pipeline is depicted in Figure 3.1. The router uses pooled student hidden states to select a sparse teacher subset. The Byte-Trie Wasserstein Loss then computes a per-teacher loss by matching decoded token paths through a shared byte trie. Gradient-agreement refinement reweights the selected teachers by how well each teacher’s KD gradient aligns with the CE objective. The final objective combines supervised cross-entropy, this routed KD signal and two router regularization terms [36, 37, 41].

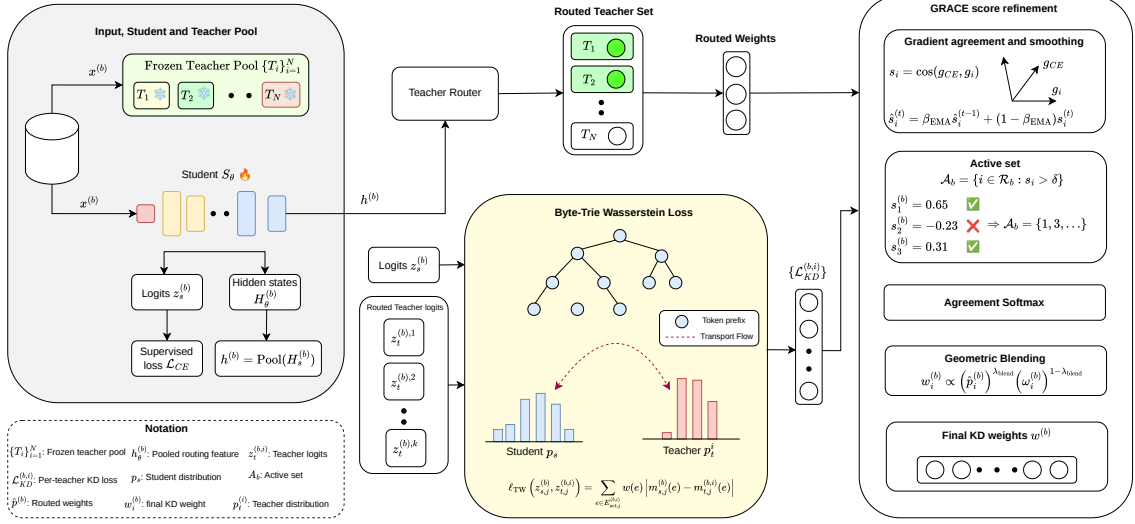


Figure 3.1: Overview of the GRACE architecture.

A GRACE update starts from a mini-batch

$$\mathcal{B} = \{(x_b, y_b)\}_{b=1}^B$$

and the frozen teacher pool $\mathcal{T}$. The student first produces logits, hidden states and the supervised cross-entropy loss for the answer tokens. The router then uses the pooled student hidden state to select a sparse routed set $\mathcal{R}_b$ for each sample. For every selected teacher, the Byte-Trie Wasserstein Loss computes a per-teacher loss $\mathcal{L}_{KD}^{(b,i)}$, comparing student and teacher distributions via the shared byte trie. Gradient-agreement refinement then adjusts routed teacher weights by comparing each teacher’s KD gradient with the supervised CE gradient. The routed KD signal for sample b is the weighted combination of teacher-specific losses:

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}.$$

The method introduces several hyperparameters (ρ_{trie} , η_{boundary} , K_{TW} , k_{route} , β_{EMA} , λ_{blend} and others). Hyperparameter values and sensitivity analysis are in Appendix B and Section 5.2. Notation is collected in Table 3.1.

Table 3.1: Notation table.

Symbol	Meaning
$\mathcal{D}$	Training set.
$\mathcal{B}$	Mini-batch.
S_θ	Trainable student VLM.
$\mathcal{T} = \{T_i\}_{i=1}^N$	Frozen teacher pool.
G_ϕ	Teacher routing network.

Symbol	Meaning
$\mathbf{z}_s^{(b)}, \mathbf{z}_t^{(b,i)}$	Student and teacher logits.
$p_{s,j}^{(b)}, p_{t,j}^{(b,i)}$	Softened token distributions.
Ω_b^{CE}	Supervised CE positions.
$\Omega_{b,i}^{KD}$	Matched KD positions.
$\mathbf{H}_s$	Student hidden states.
$\mathbf{h}$	Pooled routing feature.
ρ_{trie}	Byte-trie edge decay.
η_{boundary}	Boundary-edge weight.
K_{TW}	Sparse support size.
e_{tail}	Residual tail edge.
$\mathcal{K}_{s,j}^{(b)}, \mathcal{K}_{t,j}^{(b,i)}$	Retained top- K_{TW} token sets.
$r_{s,j}^{(b)}, r_{t,j}^{(b,i)}$	Residual tail masses.
$m_{s,j}^{(b)}(e), m_{t,j}^{(b,i)}(e)$	Sparse subtree masses.
$q_{s,j}^{(b)}$	Student non-text mass.
λ_{nt}	Non-text penalty weight.
ℓ_{TW}	Token-level Byte-Trie Wasserstein loss.
k_{route}	Number of routed teachers.
$\mathbf{r}, \mathbf{p}$	Router logits and soft router weights.
$\mathcal{R}_b$	Routed teacher set.
$\hat{\mathbf{p}}^{(b)}$	Renormalized routed weights.
$s_i, \hat{s}_i$	Raw and EMA-smoothed agreement scores.
$\mathbf{g}_{CE}, \mathbf{g}_i$	CE and teacher-specific KD gradients.
δ	Agreement threshold.
$\mathcal{A}_b$	Active routed teacher set.
$\omega^{(b)}$	Score-softmax weights.
$\mathbf{w}^{(b)}$	Final KD weights.
$\beta_{\text{EMA}}, \beta_{\text{score}}$	EMA decay and score-softmax sharpness.
$\mathcal{L}_{CE}$	Supervised CE loss.
$\mathcal{L}_{KD}^{(b,i)}$	Per-teacher KD loss.
$\bar{\mathcal{L}}_{KD}^{(b)}$	Sample-level routed KD loss.
$\mathcal{L}$	Final objective.
$\mathcal{L}_{\text{ent}}, \mathcal{L}_z$	Router regularizers.
$\alpha_{\text{KD}}, \lambda_{\text{ent}}, \lambda_z$	Final-objective weights.

3.2 Byte-Trie Wasserstein Distillation

The first obstacle in cross-tokenizer KD is that token indices are not comparable across different vocabularies: index j in one model may decode to an unrelated string in another. The Byte-Trie Wasserstein loss removes this assumption by building a shared UTF-8 byte trie over all tokenizer vocabularies and measuring transport cost through **byte-path proximity**.

For sample b and teacher T_i , GRACE restricts KD to matched answer positions $\Omega_{b,i}^{KD}$, after masking ignored labels and truncating student and teacher answer spans to a common

length. Restricting to answer positions aligns the distillation span with the supervised CE objective, which also acts only on answer tokens. Applying KD across all sequence positions would push the student’s distribution on image and question tokens that it is not trained to predict, introducing gradient noise outside the supervised span. At each position $j \in \Omega_{b,i}^{KD}$, the student and teacher logits define softened distributions

$$p_{s,j}^{(b)} = \text{softmax}\left(z_{s,j}^{(b)} / \tau_s\right), \quad p_{t,j}^{(b,i)} = \text{softmax}\left(z_{t,j}^{(b,i)} / \tau_t\right).$$

Dividing by $\tau > 1$ spreads probability mass across the vocabulary, revealing the teacher’s relative confidence over near-correct alternatives rather than collapsing it onto the single argmax token. A token the teacher ranks second or third with non-trivial probability signals that it is a plausible alternative, providing a richer gradient signal than a one-hot target would. The computation proceeds in three steps. It precomputes *canonical byte paths* for all vocabulary tokens in a shared trie, retains *sparse lexical supports* at each answer position and computes the tree-Wasserstein loss from per-edge subtree-mass imbalances. Visual placeholders and control tokens are excluded from the trie comparison and handled through a separate non-text penalty.

3.2.1 Shared Byte Trie

Applying KD by token index requires aligned vocabularies, but with different tokenizers the same index may correspond to unrelated decoded strings. GRACE therefore builds one **UTF-8 byte trie** shared by the teacher and student tokenizers. Each token piece is decoded, normalized and split into *lexical content* and *boundary metadata*. The lexical content determines the main byte path, while whitespace, continuation and end-word markers are represented as low-cost boundary information rather than as separate root branches.

This **canonicalization** suppresses tokenizer artifacts during trie-based matching: tokens with the same decoded content share byte-prefix edges even when their tokenizers use different whitespace or continuation conventions. An **EOS sentinel** is appended to each content path, which prevents prefix collapse by making a complete token such as **doc** distinct from the prefix of **document**. When boundary metadata is present, it is attached after the terminal content path, so it can affect the cost without preventing lexical prefixes from matching. Content and EOS edges receive weights that decay with depth, while optional boundary edges receive a small fixed weight:

$$w(e) = \begin{cases} \rho_{\text{trie}}^{d(e)-1}, & e \in E_{\text{content}} \cup E_{\text{EOS}}, \\ \eta_{\text{boundary}}, & e \in E_{\text{boundary}}. \end{cases} \quad (3.1)$$

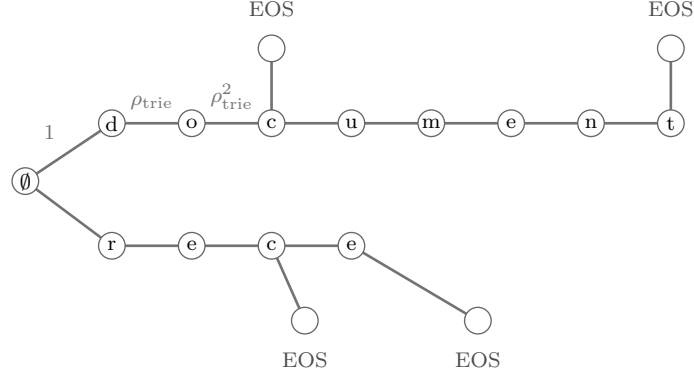


Figure 3.2: Shared byte-trie tokenizer matching.

where $0 < \rho_{\text{trie}} < 1$, $\eta_{\text{boundary}} \geq 0$ and $d(e)$ is the edge depth. The decay encodes the intuition that early divergence carries more semantic weight: two tokens that differ at depth 1 share no bytes and are completely unrelated, while two tokens that diverge only at depth 5 share a long prefix and are morphological variants of the same word. Since d_T sums only the symmetric-difference edges, depth-decayed weights make this distinction precise because early-diverging pairs accumulate more total edge weight than late-diverging ones. The trie-induced distance between two tokens u and v is the total weight of the edges on exactly one of their paths:

$$d_T(u, v) = \sum_{e \in \pi(u) \triangle \pi(v)} w(e), \quad (3.2)$$

where $\pi(u) \triangle \pi(v)$ is the symmetric difference of the two edge sets. Tokens that share a long byte prefix have small d_T , while tokens with no shared prefix pay the full path weight. This metric is the ground cost that the Wasserstein objective minimises, making it explicit why **doc** and **document** are closer than **doc** and **xyz** regardless of which tokenizer produced them [31, 33].

The trie in Figure 3.2 contains four tokens: **doc**, **document**, **rece** and **receipt**. The two document-related tokens share the path **d**→**o**→**c** before branching: **doc** terminates at the EOS node after **c**, while **document** continues through five additional edges to its own EOS. The two receipt-related tokens share **r**→**e**→**c**→**e** before branching similarly. Edge weights on content edges decay as $1, \rho_{\text{trie}}, \rho_{\text{trie}}^2, \dots$ with depth, so the transport cost between any two tokens is proportional to the length of the byte path they do *not* share: tokens with long common prefixes incur low transport cost regardless of which tokenizer produced them.

3.2.2 Sparse Support and Residual Tail Edge

The shared trie is built once from the tokenizer vocabularies, but evaluating every token path at every answer position is computationally prohibitive. GRACE therefore keeps

only the top- K_{TW} lexical student and teacher tokens after masking ignored entries. The retained tokens reuse their canonical trie paths, while unretained lexical probability mass is collapsed into a **residual tail edge** e_{tail} . Non-text visual or control tokens are not forced into the byte trie. Their probability mass is handled outside the tree comparison so the transport geometry remains lexical.

Let K_{TW} denote the sparse trie support size. Let $\mathcal{V}_{s,\text{text}}^*$ and $\mathcal{V}_{t,\text{text}}^*$ denote the non-ignored student and teacher tokens that have lexical byte paths. Ignored entries are excluded from the softmax normalizer before top- K_{TW} selection. Let $\mathcal{K}_{s,j}^{(b)} \subset \mathcal{V}_{s,\text{text}}^*$ and $\mathcal{K}_{t,j}^{(b,i)} \subset \mathcal{V}_{t,\text{text}}^*$ be the retained token sets. The residual lexical mass assigned to the tail edge is

$$r_{s,j}^{(b)} = \sum_{v \in \mathcal{V}_{s,\text{text}}^* \setminus \mathcal{K}_{s,j}^{(b)}} p_{s,j}^{(b)}(v), \quad r_{t,j}^{(b,i)} = \sum_{v \in \mathcal{V}_{t,\text{text}}^* \setminus \mathcal{K}_{t,j}^{(b,i)}} p_{t,j}^{(b,i)}(v). \quad (3.3)$$

Let $q_{s,j}^{(b)}$ denote the student probability assigned to non-text tokens. For the student distribution,

$$\sum_{v \in \mathcal{K}_{s,j}^{(b)}} p_{s,j}^{(b)}(v) + r_{s,j}^{(b)} + q_{s,j}^{(b)} = 1,$$

with the analogous identity for the teacher. Let $\pi(v)$ be the trie path for token v , including canonical content edges, the EOS edge and any optional boundary edge. Let e_{tail} be the global residual tail edge with $w(e_{\text{tail}}) = 1$. For any active trie edge e , the sparse subtree masses are

$$m_{s,j}^{(b)}(e) = \sum_{\substack{v \in \mathcal{K}_{s,j}^{(b)} \\ e \in \pi(v)}} p_{s,j}^{(b)}(v) + \mathbf{1}[e = e_{\text{tail}}] r_{s,j}^{(b)}, \quad (3.4)$$

$$m_{t,j}^{(b,i)}(e) = \sum_{\substack{v \in \mathcal{K}_{t,j}^{(b,i)} \\ e \in \pi(v)}} p_{t,j}^{(b,i)}(v) + \mathbf{1}[e = e_{\text{tail}}] r_{t,j}^{(b,i)}. \quad (3.5)$$

The three-step sparse construction (Figure 3.3) proceeds left to right. The left panel shows the student distribution: the top- K_{TW} lexical tokens are retained and the remaining lexical mass is collapsed into the residual edge. The centre panel shows the same operation for the teacher. The right panel shows the per-edge subtree masses that result: the student (blue bars) and teacher (orange bars) masses are compared at each active edge, while the green line shows the corresponding edge weight $w(e)$, which decays with depth for content edges and returns to its maximum at the residual tail edge.

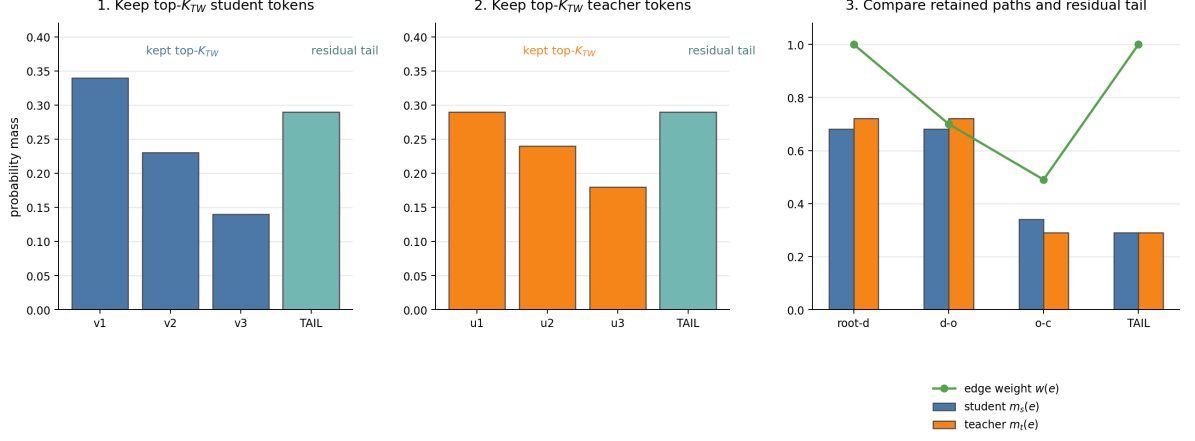


Figure 3.3: Sparse Byte-Trie Wasserstein support.

3.2.3 Byte-Trie Wasserstein Objective

With the trie structure and sparse supports defined, the loss is computed as a weighted sum of per-edge subtree-mass imbalances. Writing the dense case first makes the formula clearest: let $\mathcal{V}_s$ and $\mathcal{V}_t$ denote the full student and teacher vocabularies at a given answer position and let $m_s(e)$ and $m_t(e)$ measure how much probability each distribution routes through edge e :

$$m_s(e) = \sum_{\substack{v \in \mathcal{V}_s \\ e \in \pi(v)}} p_s(v), \quad m_t(e) = \sum_{\substack{u \in \mathcal{V}_t \\ e \in \pi(u)}} p_t(u).$$

On a tree metric, the Wasserstein distance has the closed form

$$W_T(p_s, p_t) = \sum_{e \in E} w(e) |m_s(e) - m_t(e)|. \quad (3.6)$$

This form follows from the standard W_1 Kantorovich optimal transport problem under the trie-induced tree metric [31, 33]. The full proof is provided in Appendix A. In training, GRACE evaluates the same per-edge formula on the sparse active edge set defined above, using the retained token paths and residual tail edge. The token-level Byte-Trie Wasserstein loss is

$$\ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}) = \sum_{e \in \mathcal{E}_{act,j}^{(b,i)}} w(e) |m_{s,j}^{(b)}(e) - m_{t,j}^{(b,i)}(e)| + \lambda_{nt} q_{s,j}^{(b)}, \quad (3.7)$$

where $\mathcal{E}_{act,j}^{(b,i)}$ contains the active canonical content edges, EOS edges, optional boundary edges and the residual tail edge from the student and teacher supports. The **non-text penalty** with $\lambda_{nt} \geq 0$ discourages probability mass on visual or control tokens without assigning those tokens a lexical byte path.

The per-sample, per-teacher KD loss is the mean over supervised answer tokens:

$$\mathcal{L}_{KD}^{(b,i)} = \frac{1}{|\Omega_{b,i}^{KD}|} \sum_{j \in \Omega_{b,i}^{KD}} \ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}). \quad (3.8)$$

The gradient of the trie term with respect to the student probability on retained token v at position j is

$$\frac{\partial \ell_{TW}}{\partial p_{s,j}^{(b)}(v)} = \sum_{e \in \pi(v)} w(e) \cdot \text{sign}\left(m_{s,j}^{(b)}(e) - m_{t,j}^{(b,i)}(e)\right), \quad v \in \mathcal{K}_{s,j}^{(b)}. \quad (3.9)$$

The gradient for token v accumulates signed corrections along every edge on its byte path. Tokens sharing a byte prefix with the teacher’s high-probability region therefore receive correlated gradient updates, nudging the student toward semantically related strings. This **prefix-coupling** is absent from rank-based losses such as ULD, which updates each rank position independently of decoded string content. The tree term itself remains exact Wasserstein transport under the trie metric, while the non-text penalty is additive and independent of that geometry.

Complexity Analysis. The shared byte trie is built once with cost $O(V_s \bar{L}_s + V_t \bar{L}_t)$, where V_s, V_t are the student and teacher vocabulary sizes and $\bar{L}_s, \bar{L}_t$ are average trie-path lengths. During training, each aligned answer position keeps only the top- K_{TW} lexical student and teacher tokens and accumulates probability mass along their active trie paths. The per-position cost is

$$O((V_s + V_t) \log K_{TW} + K_{TW}(\bar{L}_s + \bar{L}_t) + A),$$

where A is the number of active trie edges. Thus, sparse Byte-Trie Wasserstein matching avoids materializing the dense $|V_s| \times |V_t|$ optimal-transport matrix. The approximation error relative to the dense loss is bounded by the total unretained mass: since all edge weights satisfy $w(e) \leq 1$ and reassigning unretained mass to the tail edge changes each edge contribution by at most one unit,

$$|\ell_{TW}^{\text{sparse}} - \ell_{TW}^{\text{dense}}| \leq r_{s,j}^{(b)} + r_{t,j}^{(b,i)}. \quad (3.10)$$

The bound becomes small when K_{TW} is large enough that the tail masses are small and it vanishes when the retained set covers the full vocabulary.

3.3 Teacher Routing Network

With the cross-tokenizer loss defined, GRACE next decides which teachers to activate for each training sample. Uniform teacher averaging ignores how each teacher performs across

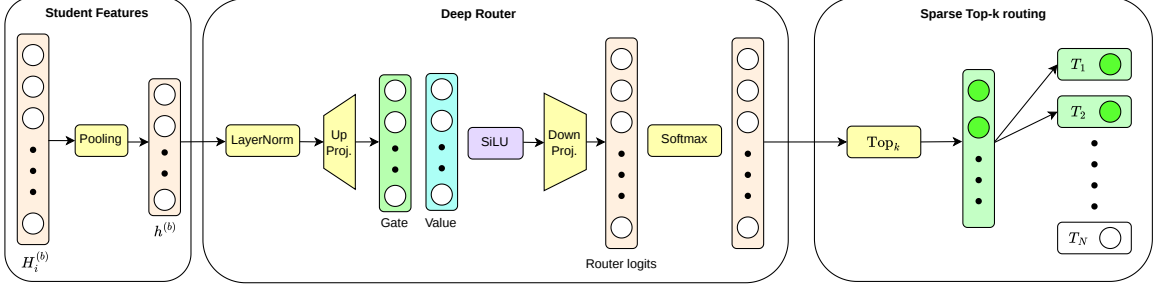


Figure 3.4: Teacher routing network.

different samples. GRACE therefore uses a **student-conditioned router** to propose a small teacher set per sample. Figure 3.4 shows the full routing pipeline [21, 36, 37].

Let $\mathbf{H}_s^{(b)}$ be the student hidden states before the language-model head. GRACE pools the supervised answer positions to obtain the routing feature

$$\mathbf{h}^{(b)} = \text{Pool}(\mathbf{H}_s^{(b)}).$$

Answer-position pooling restricts the routing feature to the supervised span, so the router conditions on a representation that mixes visual context, question text and answer content, which jointly determine which teacher is most useful for a given sample.

A learned **MLP router** $G_\phi(\cdot)$ maps this feature to teacher logits and soft weights:

$$\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)}), \quad \mathbf{p}^{(b)} = \text{softmax}(\mathbf{r}^{(b)}).$$

Concretely, G_ϕ is a **gated MLP** with a SwiGLU-style block. The hidden projection is split into a *value* branch and a *gate* branch and their element-wise product $\text{SiLU}(\mathbf{g}) \odot \mathbf{v}$ forms the routing representation before the final linear head. A plain linear projection cannot capture the non-linear feature interactions that determine teacher usefulness because whether a particular teacher is the best supervisor depends on joint patterns in $\mathbf{h}^{(b)}$, such as the image being chart-heavy *and* the question requesting a numerical value, not on either cue alone. The gating mechanism addresses this directly: one branch creates a rich intermediate representation from the student hidden state and the other branch learns which dimensions of that representation are actually relevant for the routing decision, suppressing irrelevant features rather than propagating them. The router then keeps the top- k_{route} teachers:

$$\mathcal{R}_b = \text{TopK}(\mathbf{r}^{(b)}, \min(k_{\text{route}}, N)).$$

Hard selection is used rather than a soft threshold because keeping low-weight teachers still requires computing the full Byte-Trie Wasserstein loss for each one. Eliminating them entirely removes their noise from the distillation signal rather than merely downweighting

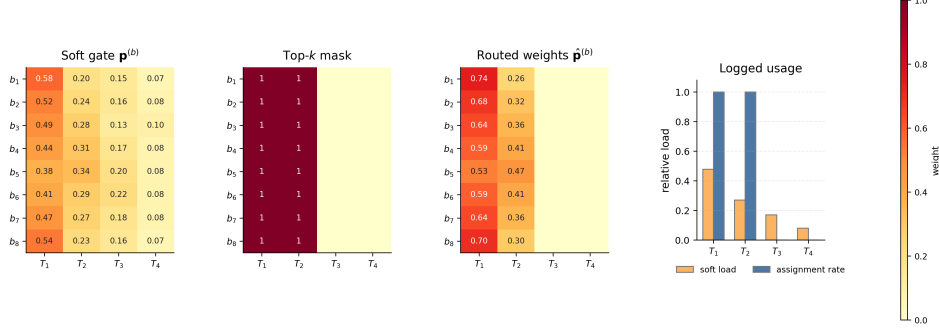


Figure 3.5: Top- k routing dynamics.

it. Weights outside $\mathcal{R}_b$ are zeroed and the rest renormalized:

$$\hat{p}_i^{(b)} = \frac{\mathbf{1}[i \in \mathcal{R}_b] p_i^{(b)}}{\sum_{j=1}^N \mathbf{1}[j \in \mathcal{R}_b] p_j^{(b)}}. \quad (3.11)$$

Because $\mathbf{h}^{(b)}$ reflects the current student hidden states, the routed set $\mathcal{R}_b$ changes as the student evolves, so teacher selection adapts to both input content and the current student state. Top- k routing also limits the active teacher set to k_{route} candidates and separates candidate selection from final weighting: the router proposes teachers by input relevance and Section 3.4 adjusts their weights by gradient compatibility.

The effect of top- k routing across eight samples and four teachers is visible in Figure 3.5. In the leftmost heatmap, T_1 consistently receives the highest soft gate weight, followed by T_2 , while T_3 and T_4 remain low throughout. The top- k mask (second panel) retains only T_1 and T_2 for all samples, zeroing the rest. After renormalization, the routed weights $\hat{\mathbf{p}}^{(b)}$ (third panel) are split between T_1 and T_2 , with T_1 holding the majority. The bar chart on the right confirms that T_1 and T_2 capture all routing assignments in this mini-batch, while T_3 and T_4 receive none.

Hard top- k selection can cause the router to **collapse onto a single teacher** if left unconstrained, so GRACE uses entropy and z -loss stabilizers. The entropy term discourages premature **routing collapse** [36, 37]:

$$\mathcal{L}_{ent} = -\frac{1}{B} \sum_{b=1}^B H(\mathbf{p}^{(b)}). \quad (3.12)$$

The z -loss penalizes large router logit magnitudes:

$$\mathcal{L}_z = \frac{1}{B} \sum_{b=1}^B \left(\log \sum_{i=1}^N e^{r_i^{(b)}} \right)^2. \quad (3.13)$$

These auxiliary losses stabilize teacher selection, while CE and routed KD remain the main training signals.

3.4 Gradient-Agreement Refinement

Routing proposes teachers relevant to each sample, but a selected teacher may still produce a KD gradient that conflicts with the supervised CE objective. GRACE computes one agreement score per teacher per mini-batch and uses it to reweight each sample’s routed teachers.

3.4.1 Agreement Score Computation

Let $\mathbf{g}_{CE} = \nabla_{\Theta} \mathcal{L}_{CE}$ be the supervised gradient over trainable student parameters Θ . For teacher T_i , GRACE averages its KD loss over the mini-batch to get

$$\mathbf{g}_i = \nabla_{\Theta} \left(\frac{1}{B} \sum_{b=1}^B \mathcal{L}_{KD}^{(b,i)} \right),$$

where $\mathcal{L}_{KD}^{(b,i)}$ is the Byte-Trie Wasserstein loss from Equation 3.8. The raw **agreement score** is the cosine similarity

$$s_i = \frac{\langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle}{\max(\|\mathbf{g}_{CE}\|_2 \|\mathbf{g}_i\|_2, \epsilon_{\text{num}})}. \quad (3.14)$$

Cosine similarity is well-suited to this comparison: both losses optimize the same parameters and the normalization removes teacher-to-teacher scale differences. This is clearest from a first-order expansion of one gradient step with routed KD:

$$\Delta\theta = -\eta \left(\mathbf{g}_{CE} + \alpha_{\text{KD}} \sum_i w_i \mathbf{g}_i \right).$$

The first-order change in the supervised CE loss is

$$\Delta \mathcal{L}_{CE} \approx \mathbf{g}_{CE}^{\top} \Delta\theta = -\eta \|\mathbf{g}_{CE}\|_2^2 - \eta \alpha_{\text{KD}} \sum_i w_i \langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle.$$

When $\langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle > 0$, teacher T_i ’s KD direction **supports** the supervised CE update. When the inner product is negative, the teacher **pushes against** the supervised CE objective. Under the simplex constraints $w_i \geq 0$ and $\sum_i w_i = 1$, the unregularized first-order objective would place all KD weight on the routed teacher with the largest inner product $\langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle$. GRACE uses cosine agreement instead, comparing **gradient directions and not magnitudes**. The score-softmax gives a smooth relaxation of the hard winner-take-all selection. Gradient agreement thus serves both roles: **filtering conflicting teachers** and **up-weighting** those whose KD direction aligns with the CE update [42]. Each s_i is computed once per mini-batch at the teacher level and per-sample variation arises through intersection with each sample’s routed set $\mathcal{R}_b$.

3.4.2 Warmup, EMA Smoothing and Active Set

During **warmup** (the first $\rho_g T_{\text{train}}$ steps), routed weights are used unchanged: $w_i^{(b)} = \hat{p}_i^{(b)}$. Agreement scoring is skipped early because an unstable student produces scores too noisy to be useful for filtering. After warmup, the raw score is first thresholded:

$$\tilde{a}_i = \mathbf{1}[s_i > \delta],$$

then smoothed with **EMA**:

$$\hat{s}_i = \beta_{\text{EMA}} \hat{s}_i^{\text{prev}} + (1 - \beta_{\text{EMA}}) s_i.$$

A single mini-batch score is too noisy to be a reliable filter: a teacher that is consistently aligned may produce a negative score on one batch purely due to sampling variation. EMA accumulates a smoothed signal across steps so that filtering reflects a teacher’s sustained directional tendency rather than reacting to individual batch fluctuations. In short, s_i captures the current-batch gradient direction and drives the threshold test, while $\hat{s}_i$ accumulates a smoothed estimate across steps and drives the final weighting. Figure 3.6 illustrates both effects. The left panel shows that the per-batch score (orange) oscillates widely across training steps, while the EMA (blue) tracks a smooth, slowly-moving trend. The centre panel compares the current s_i and smoothed $\hat{s}_i$ across four teachers for one step: T_4 has a near-zero or negative current score that would suppress it if used directly, but its smoothed score remains slightly positive, reflecting recent history. The right panel shows the consequence for weighting: using raw batch scores concentrates almost all gradient weight on a single teacher, while using smoothed scores distributes weight more evenly across the active set, preventing a single noisy batch from dominating teacher selection.

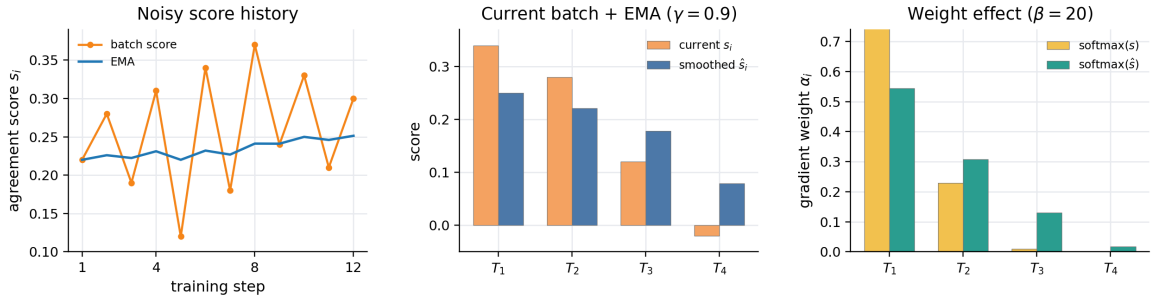


Figure 3.6: EMA smoothing.

With smoothed scores in hand, GRACE constructs a **per-sample active set** by intersecting the threshold flag $\tilde{a}_i$ with the routed set:

$$\mathcal{A}_b = \{i \in \mathcal{R}_b : \tilde{a}_i = 1\}. \quad (3.15)$$

Finally, the smoothed scores pass a softmax over $\mathcal{A}_b$ to produce agreement weights ω :

$$\omega_i^{(b)} = \frac{\exp(\beta_{\text{score}} \hat{s}_i) \mathbf{1}[i \in \mathcal{A}_b]}{\sum_{j=1}^N \exp(\beta_{\text{score}} \hat{s}_j) \mathbf{1}[j \in \mathcal{A}_b]}, \quad (3.16)$$

These agreement weights $\omega_i^{(b)}$ must then be combined with the router’s relevance scores $\hat{p}_i^{(b)}$. Three edge cases, an empty active set, nearly tied scores and a clear winner, each call for a different rule.

3.4.3 Fallback, Uniform and Blended Weighting

The final rule, summarized in Figure 3.7, chooses among three cases, each handling a failure mode the others cannot. If $\mathcal{A}_b$ is empty, every selected teacher is conflicting and there is no agreement signal to use. GRACE **falls back** to the routed weights $\hat{p}^{(b)}$ because they still reflect input-level relevance and training must continue. If the spread between the maximum and minimum agreement scores inside $\mathcal{A}_b$ is below ϵ_{fb} (so the active teachers are nearly tied), acting on small score differences would amplify noise into meaningless weight differences. GRACE therefore uses **uniform weights** over $\mathcal{A}_b$ as the more robust choice. Otherwise, when the agreement signal is strong enough to differentiate teachers, it **geometrically blends** the routed weights $\hat{p}$ and the agreement weights ω :

$$w_i^{(b)} = \frac{(\hat{p}_i^{(b)})^{\lambda_{\text{blend}}} (\omega_i^{(b)})^{1-\lambda_{\text{blend}}} \mathbf{1}[i \in \mathcal{A}_b]}{\sum_{j=1}^N (\hat{p}_j^{(b)})^{\lambda_{\text{blend}}} (\omega_j^{(b)})^{1-\lambda_{\text{blend}}} \mathbf{1}[j \in \mathcal{A}_b]}. \quad (3.17)$$

Here, $\lambda_{\text{blend}} \in [0, 1]$ interpolates between pure score weighting and pure router weighting. The geometric form is deliberate: because the two terms are multiplied rather than added, a teacher with near-zero weight under either signal is suppressed in the final weight regardless of how high the other signal is. An arithmetic blend would allow a highly gradient-aligned teacher to be included despite low routing relevance, or vice versa. The geometric blend requires both signals to agree, with λ_{blend} controlling how much each signal contributes to that joint decision.

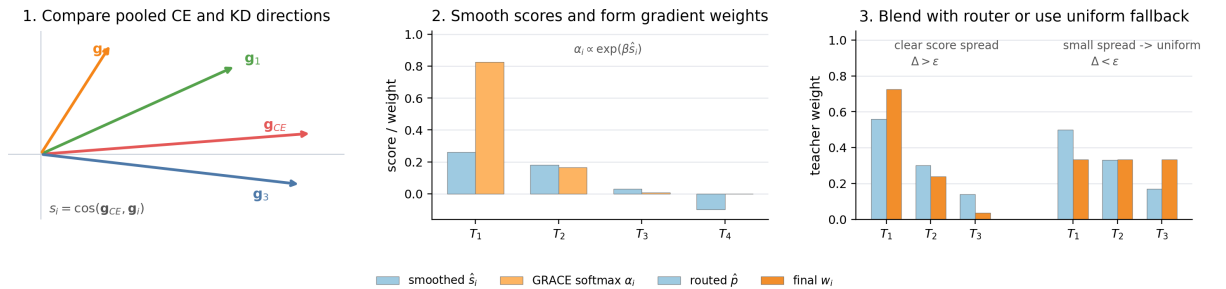


Figure 3.7: GRACE score refinement.

The three panels of Figure 3.7 step through the refinement pipeline in order. On the left, the gradient-direction diagram shows $\mathbf{g}_1$ (green) pointing in nearly the same direction as $\mathbf{g}_{CE}$ (red), $\mathbf{g}_f$ (orange) diverging at a moderate angle and $\mathbf{g}_3$ (blue) pointing in the opposite direction, indicating conflict with the supervised update. The centre panel shows the consequence for weights. The teacher with the highest smoothed score receives a disproportionately large softmax weight, while the teacher whose score falls below the threshold is excluded from the active set entirely. On the right, the two non-fallback outcomes are contrasted: when the score spread exceeds ϵ_{fb} , the geometric blend shifts the final weight toward the gradient-preferred teacher. When the spread is below the threshold, all active teachers receive equal final weights.

Each of the three panels in Figure 3.8 corresponds to one of the weighting rules. When gradient agreement strongly prefers one teacher (left panel, large Δ), the final blended weight for that teacher is pulled well above its routed weight. When the router and gradient agreement favor different teachers (centre panel, moderate Δ), the blend preserves meaningful contributions from both. When the score spread is negligible (right panel, $\Delta < \epsilon$), GRACE treats the active teachers as indistinguishable and assigns them equal weight.

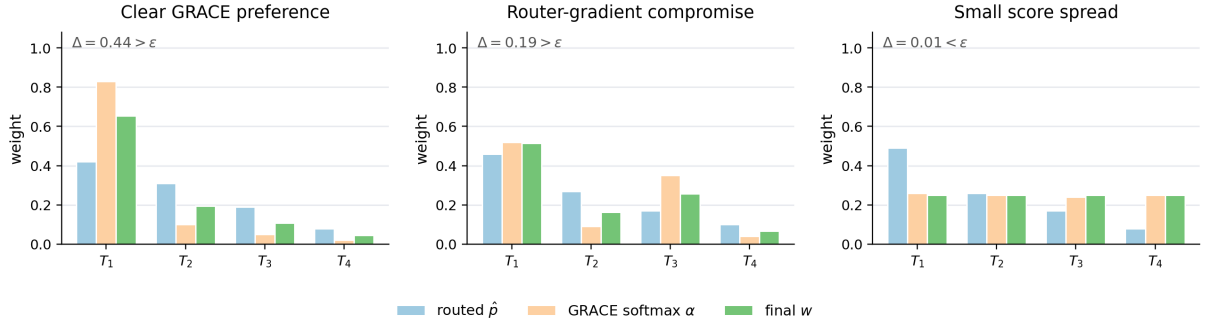


Figure 3.8: Representative GRACE weighting cases.

Following a single sample through all four stages (Figure 3.9): the dense gate softmax $\mathbf{p}$ assigns nonzero probability to all four teachers. Top- k selection then retains only T_1 and T_2 and renormalizes their weights. The agreement softmax ω (third panel) amplifies T_1 further based on its higher agreement score, leaving T_2 with a smaller share. The geometric blend in the final panel produces a weight that lies between the router preference and the agreement preference, with neither signal dominating entirely.

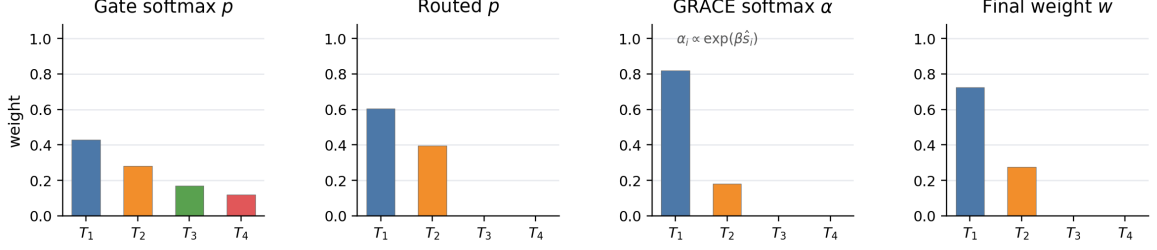


Figure 3.9: Sample-wise weight refinement.

3.5 Routed KD Aggregation and Final Objective

The three components described in the preceding sections, Byte-Trie Wasserstein matching, sparse routing and gradient-agreement refinement, each produce a per-teacher, per-sample quantity: a loss value $\mathcal{L}_{KD}^{(b,i)}$ and a final weight $w_i^{(b)}$. Gradient-agreement refinement restricts these weights to the routed set $\mathcal{R}_b$, so each sample draws on at most k_{route} teachers. This section combines both quantities into a single training objective whose key property is that $w_i^{(b)}$ is **sample-specific**, allowing different examples in the same mini-batch to draw on different teachers.

At the sample level, only teachers in the routed set $\mathcal{R}_b$ contribute to the KD loss:

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}, \quad (3.18)$$

where $w_i^{(b)}$ reflects whichever of the three weighting rules applies (fallback, uniform or blended). The mini-batch KD term averages these:

$$\mathcal{L}_{KD} = B^{-1} \sum_{b=1}^B \bar{\mathcal{L}}_{KD}^{(b)}.$$

Because $w_i^{(b)}$ is computed per sample, different examples in the same mini-batch can emphasize different teachers. Figure 3.10 shows this for text-heavy, chart-heavy and ambiguous samples.

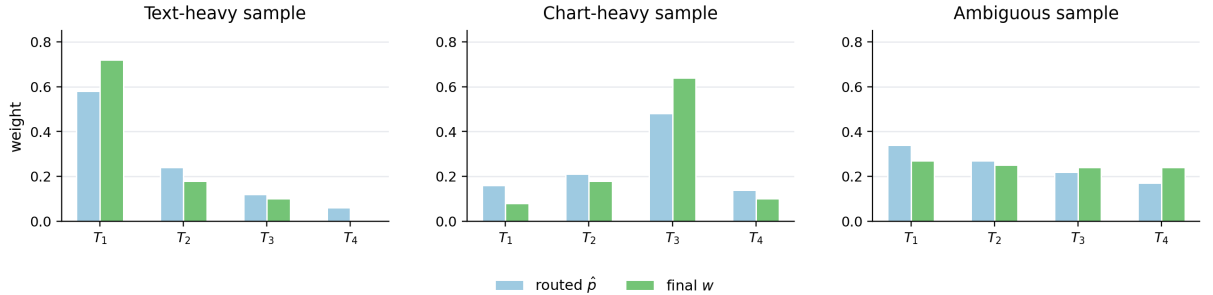


Figure 3.10: Sample-specific weighting.

Each panel compares the routed weight $\hat{p}$ (blue) with the final weight w (green) across four teachers. For the text-heavy sample, T_1 is already the top router choice and gradient agreement amplifies it further, while the lower-ranked teachers are suppressed. For the chart-heavy sample, T_3 is preferred by the router and agreement reinforces this, shifting the dominant teacher away from T_1 . For the ambiguous sample, the router weights are spread across all four teachers and the agreement scores provide no strong signal, so the final weights remain close to the routed weights with no single teacher dominating.

The complete GRACE objective is

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{KD}\mathcal{L}_{KD} + \lambda_{ent}\mathcal{L}_{ent} + \lambda_z\mathcal{L}_z, \quad (3.19)$$

where $\mathcal{L}_{CE}$ anchors the student to supervised answer tokens, $\mathcal{L}_{KD}$ transfers routed teacher supervision and $\mathcal{L}_{ent}$ together with $\mathcal{L}_z$ regularize the router, while teacher parameters remain frozen throughout. Together, the four terms directly address the three constraints from the opening of this chapter: $\mathcal{L}_{KD}$ handles **tokenizer mismatch** through the byte-trie loss, the routing weights inside it handle **per-sample teacher relevance** and the gradient-agreement scores that set those weights handle **optimization conflict**.

Algorithm 1 One GRACE training step

Require: Mini-batch $\mathcal{B} = \{(x_b, y_b)\}_{b=1}^B$, frozen teachers $\{T_i\}_{i=1}^N$, student S_θ , router G_ϕ

- 1: Run the student on $\mathcal{B}$ to obtain logits $\mathbf{z}_s^{(b)}$, hidden states $\mathbf{H}_s^{(b)}$ and supervised loss $\mathcal{L}_{CE}$
- 2: **for** each sample b **do**
- 3: Pool hidden states at answer positions to form $\mathbf{h}^{(b)}$
- 4: Compute router logits $\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)})$ and probabilities $\mathbf{p}^{(b)} = \text{softmax}(\mathbf{r}^{(b)})$
- 5: Select routed set $\mathcal{R}_b = \text{TopK}(\mathbf{r}^{(b)}, \min(k_{\text{route}}, N))$
- 6: Renormalize routed weights to obtain $\hat{\mathbf{p}}^{(b)}$
- 7: **end for**
- 8: **for** each teacher T_i selected by at least one sample **do**
- 9: Load cached teacher logits $\mathbf{z}_t^{(b,i)}$
- 10: Compute $\mathcal{L}_{KD}^{(b,i)}$ with Byte-Trie Wasserstein matching on aligned answer positions
- 11: Compute teacher-level gradient $\mathbf{g}_i$ and agreement score s_i
- 12: **end for**
- 13: Smooth agreement scores with EMA and construct active sets $\mathcal{A}_b$
- 14: Compute final teacher weights $w_i^{(b)}$ using fallback, uniform or geometric blend rules
- 15: Aggregate routed KD for each sample

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}$$

16: Optimize the final objective

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{\text{KD}} \frac{1}{B} \sum_{b=1}^B \bar{\mathcal{L}}_{KD}^{(b)} + \lambda_{\text{ent}} \mathcal{L}_{\text{ent}} + \lambda_z \mathcal{L}_z$$

Chapter 4

Evaluation Methodology

This chapter establishes the experimental setting that makes the results in Chapter 5 interpretable. It describes the student and teacher models, the three VQA benchmarks and their evaluation metrics, the baselines that GRACE is compared against and the ablation variants used to isolate each component’s contribution. Readers who want to reproduce or extend the experiments will find the complete hyperparameter values in Appendix B.

4.1 Experimental Setup

The student models are **SmolVLM-500M** and **SmolVLM-256M** [19], evaluated with **VLM-Eval-Kit** [44], which provides a unified inference and evaluation pipeline across all three benchmarks. For each sample, the toolkit loads the image-question pair, applies dataset-specific prompts and runs the model to produce a text answer.

Before training, all benchmark samples are converted into a unified **LLaVA-style multimodal instruction format** [3] as shown in listing 4.1, so all three benchmarks share the same data loader and training pipeline.

```
1 {  
2   "image": "sample_000123.jpg",  
3   "conversations": [  
4     {"from": "human", "value": "<image>\nWhat is the invoice number?"},  
5     {"from": "gpt", "value": "4572"}  
6   ]  
7 }
```

Listing 4.1: Representative LLaVA-style sample used after dataset conversion.

We run all training jobs on a single high-memory **NVIDIA H200** GPU or a comparable single-GPU **Vast.ai** instance. Distillation training uses **DeepSpeed ZeRO Stage 2**, **bf16** mixed precision, **TF32** matrix multiplication and gradient checkpointing to control memory, keeping the student trainable and all teachers frozen [45, 46]. Full optimization and routing parameter values are reported in Appendix B, Table B.1. Each configuration was run once because the per-run compute cost made multiple seeds impractical,



Figure 4.1: Representative dataset examples.¹

with implications discussed in the Conclusion. Teacher logit storage and remote caching are in Appendix B.2 and Appendix B.3.

4.2 Datasets

We evaluate three text-centric VQA benchmarks: **TextVQA**, **DocVQA** and **ChartQA**. Together they span OCR-heavy, layout-based and numerical visual reasoning [4–6].

TextVQA tests question answering on natural images containing scene text [4]. *Sample question*: “What word is written on the stop sign?”

DocVQA tests question answering on scanned document images, where layout, paragraphs, tables and forms determine the answer [5]. *Sample question*: “What is the invoice number shown in this document?”

ChartQA tests question answering over charts, requiring numerical and logical reasoning [6]. *Sample question*: “Which year has the highest sales value in the chart?”

No single benchmark captures all three and a method that excels at only one would still fail on the others. Figure 4.1 shows representative samples from each dataset. The TextVQA panel shows a stop sign and the answer requires reading text embedded in the scene. The DocVQA panel shows a receipt with fields, item rows and totals. The ChartQA panel shows a bar chart and answering requires comparing bar heights and reading values from the plot. Table 4.1 lists the split sizes.

¹Images adapted from Wikimedia Commons sources [47–49].

Table 4.1: Dataset split sizes.

Dataset	Train	Val	Test	Total
TextVQA	34,602	5,000	5,734	45,336
DocVQA	39,463	5,349	5,188	50,000
ChartQA-H (human)	7,398	960	1,250	9,608
ChartQA-M (machine)	20,901	960	1,250	23,111
ChartQA (H+M, combined)	28,299	1,920	2,500	32,719

Each benchmark uses a distinct metric: TextVQA uses **VQA Accuracy**, DocVQA uses **ANLS** and ChartQA uses **Relaxed Accuracy**. For TextVQA, the per-question score for a predicted answer $\hat{a}$ is $s_{\text{VQA}} = \min(\# \text{annotators that provided } \hat{a}/3, 1)$ and the final score is the mean over all questions [4]. For DocVQA, the metric is [5]:

$$\text{NLS}(a, \hat{a}) = 1 - \frac{d_{\text{Lev}}(a, \hat{a})}{\max(|a|, |\hat{a}|)}, \quad s_{\text{ANLS}} = \begin{cases} \text{NLS}(a, \hat{a}), & \text{NLS} \geq 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

For ChartQA, we report Relaxed Accuracy [6]. Text predictions must match exactly, while numeric predictions are scored with

$$s_{\text{ChartQA}} = \begin{cases} \mathbf{1}[\hat{y} = y], & y = 0, \\ \mathbf{1}\left[\frac{|\hat{y}-y|}{|y|} \leq 0.05\right], & y \neq 0. \end{cases}$$

4.3 Baselines

Table 4.2 organizes the compared methods into four groups: non-distillation references, same-family KD, single-teacher transfer with mismatched tokenizers and heterogeneous teacher methods. The teacher setup column references SmolVLM2 [19, 50], Qwen2-VL [8, 51], Qwen2.5-VL [7, 52], InternVL3 [53, 54] and Gemma 3 [55].

The same-family baselines use **SmolVLM2-2.2B-Instruct** as teacher for forward KL, reverse KL and Jensen-Shannon distillation, testing aligned-vocabulary transfer before any tokenizer mismatch is introduced [15, 26, 56]. The different-family single-teacher setting uses ULD with **Qwen2-VL-2B-Instruct** outside the SmolVLM family, testing transfer across mismatched tokenizers with one teacher [20]. The heterogeneous teacher setting compares uniform averaging, SHM-CKA, REINFORCE and GRACE over a mixed teacher pool, separating static averaging, hidden-state matching and adaptive teacher assignment [22, 28, 33, 43]. All trainable competitor methods retain the supervised term $\mathcal{L}_{CE}$, so

Table 4.2: Baseline groups.

Group	Methods	Teacher setup	Purpose
Non-distillation	Original, full fine-tuning	No teacher	Student reference points
Same-family	Forward KL, Reverse KL, Jensen-Shannon	SmolVLM2-2.2B-Instruct	Single-teacher KL objectives within the SmolVLM family
Cross-family single-teacher	ULD	Qwen2-VL-2B-Instruct	Cross-family transfer
Heterogeneous teacher baselines	Uniform averaging, SHM-CKA, REINFORCE, GRACE (Ours)	Mixed set from multiple VLM families	Compares static averaging, hidden-state matching and adaptive assignment

their differences come from the added transfer signal: the logits-only baselines change only the KD term, SHM-CKA adds hidden-state alignment and REINFORCE adds a policy-gradient term [21, 29, 57]. In the descriptions below, γ denotes a generic KD mixing coefficient, while α_{KD} is reserved for the GRACE objective in Equation 3.19.

Original. Direct evaluation of the released student, with no additional optimization. This measures the native capability of the compact model before any task adaptation or distillation.

Full fine-tuning. Uses $\mathcal{L} = \mathcal{L}_{CE}$ with no teacher, so it isolates the benefit of external supervision alone.

Forward KL. Uses $\mathcal{L}_{KD} = \text{KL}(p_t \| p_s)$ for same-family soft-target distillation when teacher and student vocabularies are aligned [15].

Reverse KL. Uses $\mathcal{L}_{KD} = \text{KL}(p_s \| p_t)$ [56, 58] as a student-centered divergence baseline, which tests whether matching the teacher from the student’s current distribution is suitable for VLM distillation. Because the reverse-KL gradient is weighted by p_s , this objective tends to **emphasize regions where the student already assigns probability**.

Jensen-Shannon. Uses the symmetric divergence

$$\frac{1}{2}\text{KL}(p_s \| m) + \frac{1}{2}\text{KL}(p_t \| m), \quad m = \frac{1}{2}(p_s + p_t).$$

Jensen-Shannon tests whether a bounded symmetric objective is more stable than KL [26].

ULD. Uses $\mathcal{L}_{KD} = \|\pi(p_s) - \pi(p_t)\|_1$ [20], where $\pi(\cdot)$ sorts the softened probability vector in descending order and pads unequal vocabulary sizes before comparison.

SHM-CKA. Extends ULD with a hidden-state CKA penalty [20, 28, 29]. For the heterogeneous setting, both the ULD and layer-matching terms are averaged uniformly across

teachers before combining with $\mathcal{L}_{CE}$. Full implementation details are given in Appendix B.1.

Uniform multi-teacher. Averages the heterogeneous teacher losses with fixed equal weights, $\mathcal{L}_{KD} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{KD}^{(i)}$ [21, 43], isolating the effect of teacher diversity from any adaptive assignment.

REINFORCE. Replaces deterministic routing with a stochastic policy that samples teachers and updates the selector with REINFORCE [21, 57], using policy-gradient selection in place of gradient-agreement scoring. The selector objective is

$$\mathcal{L}_{\text{pol}} = -(r - b) \log p(\mathbf{m} \mid \boldsymbol{\pi}), \quad (4.1)$$

where r is the reward, b is an EMA reward baseline and $\mathbf{m}$ is the sampled teacher mask. Its final training objective is

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{KD} + \lambda_{\pi} \mathcal{L}_{\text{pol}}. \quad (4.2)$$

Additional baseline implementation details are in Appendix B.1.

Table 4.3 reports pre-distillation scores for both student models and each teacher across the three benchmarks, evaluated using **VLM-Eval-Kit** [44]. These scores serve as reference points for computing teacher-student gaps in Chapter 5.

Table 4.3: Student and teacher model results.

Model	TextVQA	DocVQA	ChartQA
<i>Student Models</i>			
SmolVLM-500M	56.75	59.52	62.80
SmolVLM-256M	47.17	48.64	45.04
<i>Teacher Models</i>			
Qwen2.5-VL-3B-Instruct	79.30	93.90	94.24
InternVL3-1B	74.10	81.90	75.30
Gemma-3-4B-it	57.80	75.80	63.60
Qwen2-VL-2B-Instruct	79.70	90.10	89.28

Model sources are SmolVLM [19], Qwen2.5-VL [7, 52], InternVL3 [53], Gemma 3 [55] and Qwen2-VL [8, 51]. The scores reveal meaningful specialization: Qwen2-VL leads on TextVQA (79.70) and is competitive on DocVQA (90.10), while Qwen2.5-VL achieves the highest scores on both DocVQA (93.90) and ChartQA (94.24). InternVL3-1B contributes competitive but not top scores across all three benchmarks. Gemma-3-4B-it is the weakest

teacher, particularly on TextVQA (57.80), where it barely exceeds the original student baseline. **No single teacher dominates all three benchmarks**, motivating the per-sample routing approach described in Chapter 3.

4.4 Ablation Settings

The ablation study in Section 5.2 isolates the main components of GRACE on DocVQA with all other settings fixed. Each variant changes one mechanism, separating the contributions of teacher diversity, sparse routing, gradient-agreement weighting and EMA score smoothing.

Cross-family ULD. Uses one teacher outside the SmolVLM family:

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{KD} \mathcal{L}_{ULD}, \quad \mathcal{L}_{ULD} = \|\pi(p_s) - \pi(p_t)\|_1.$$

Uniform averaging. Adds the full heterogeneous teacher pool with fixed equal weights, isolating teacher diversity from adaptive weighting:

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{KD} \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{TW}^{(i)}.$$

Routing only. Keeps sparse teacher selection but replaces router weights with uniform weights within the routed set, testing top- k routing without gradient-agreement refinement:

$$w_i^{(b)} = u_i^{(b)} = \frac{\mathbf{1}[i \in \mathcal{R}_b]}{|\mathcal{R}_b|}.$$

Full GRACE. Combines trie-Wasserstein KD, sparse routing and gradient-agreement weighting:

$$w_i^{(b)} \propto (\hat{p}_i^{(b)})^{\lambda_{\text{blend}}} (\omega_i^{(b)})^{1-\lambda_{\text{blend}}}.$$

Two additional ablation variants isolate the final weighting mechanism, reported in Table 5.4. The $\lambda_{\text{blend}} = 1.0$ variant keeps the threshold on the active set and fallback rules but removes the softmax over agreement scores from the high-confidence case:

$$w_i^{(b)} \propto \hat{p}_i^{(b)} \mathbf{1}[i \in \mathcal{A}_b].$$

Unlike Routing only, which weights all routed teachers, this variant restricts weights to the active set $\mathcal{A}_b$. The $\beta_{\text{EMA}} = 0$ variant removes temporal smoothing by setting $\hat{s}_i = s_i$ before softmax weighting over agreement scores.

Chapter 5

Experiment Results

This chapter answers RQ1, RQ2 and RQ3 through main comparison results, ablation study and efficiency analysis, then discusses baseline behavior.

5.1 Main Comparison Results

Student and teacher reference scores are in Table 4.3 (Chapter 4). Using the highest teacher score per benchmark as the ceiling, the teacher-student gap is 22.95 points on TextVQA, 34.38 on DocVQA and 31.44 on ChartQA for SmolVLM-500M and correspondingly larger for SmolVLM-256M.

Table 5.1 collects non-distillation references, same-family KD with **SmolVLM2-2.2B-Instruct** [19, 50] (forward KL, reverse KL, JSD), cross-family ULD with a single teacher and multi-teacher methods (Uniform, SHM-CKA, REINFORCE and GRACE) sharing the same four-teacher pool from the Qwen, InternVL and Gemma families [7, 8, 51–53, 55]. Each cell reports the best run from our completed experiments.

Table 5.1: Main comparison across TextVQA, DocVQA and ChartQA.

Category	Method	Teacher	Student	TextVQA	DocVQA	ChartQA
Non-distillation references						
Original	None	None	SmolVLM-500M	56.75	59.52	62.80
			SmolVLM-256M	47.17	48.64	45.04
Reference	Fine-tuning	None	SmolVLM-500M	56.96	61.96	63.00
			SmolVLM-256M	49.83	42.03	49.68
Same-family distillation baselines						
Same-family	Forward KL	SmolVLM2-2.2B	SmolVLM-500M	59.00	63.55	64.52
			SmolVLM-256M	50.88	49.46	56.80
	Reverse KL		SmolVLM-500M	44.73	29.28	30.16
			SmolVLM-256M	29.75	23.28	17.20
	Jensen-Shannon		SmolVLM-500M	42.33	36.43	43.75
			SmolVLM-256M	33.84	33.12	50.93
Different-family single-teacher baselines						
Cross-family	ULD	Qwen2-VL-2B-Instruct	SmolVLM-500M	61.49	65.28	66.92
			SmolVLM-256M	53.95	52.24	60.96
Multi-teacher baselines						
Multi-teacher	Uniform	†	SmolVLM-500M	63.88	69.59	75.60
			SmolVLM-256M	54.57	55.70	63.68
	SHM-CKA		SmolVLM-500M	60.47	63.32	76.88
			SmolVLM-256M	55.93	57.24	62.40
	REINFORCE		SmolVLM-500M	63.26	64.00	70.27
			SmolVLM-256M	56.31	58.53	68.74
	GRACE (Ours)		SmolVLM-500M	69.38	73.11	80.20
			SmolVLM-256M	60.73	60.48	72.32
† Teacher pool: Qwen2-VL-2B-Instruct, Qwen2.5-VL-3B-Instruct, InternVL3-1B and Gemma-3-4B-it. Model sources are cited above.						

† Teacher pool: Qwen2-VL-2B-Instruct, Qwen2.5-VL-3B-Instruct, InternVL3-1B and Gemma-3-4B-it. Model sources are cited above.

GRACE leads on all three benchmarks for both student sizes. Fine-tuning barely improves over the original student scores and same-family logit-based KD depends strongly on the divergence: forward KL improves slightly, while reverse KL and Jensen-Shannon often reduce performance. Cross-family ULD outperforms same-family KD and uniform multi-teacher averaging improves further, showing the gain from teacher diversity. Compared with the strongest non-GRACE baseline, GRACE improves by +5.50, +3.52 and +3.32 points for SmolVLM-500M and by +4.42, +1.95 and +3.58 points for SmolVLM-256M on TextVQA, DocVQA and ChartQA respectively. REINFORCE is the closest structural competitor, since it also routes adaptively, yet GRACE leads by 5.50 points on TextVQA for SmolVLM-500M. The difference lies in the selection signal: cosine similarity between the KD and CE gradients is recomputed every mini-batch and directly measures optimization compatibility, while the REINFORCE reward is updated per evaluation checkpoint and

captures gradient alignment only indirectly. Section 5.4 examines both behaviors further.

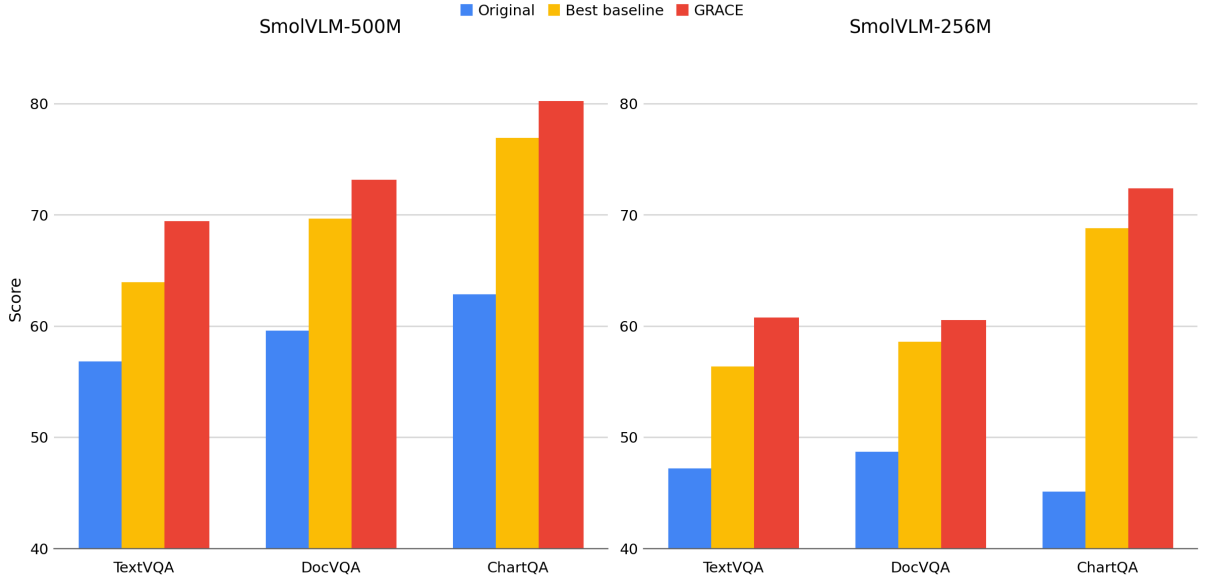


Figure 5.1: Grouped benchmark comparison.

Figure 5.1 summarises the comparison visually. Each panel groups TextVQA, DocVQA and ChartQA side by side, with three bars per benchmark: the original student (blue), the best non-GRACE baseline (orange) and GRACE (red). For SmolVLM-500M the red bar extends clearly above the orange on all three benchmarks, with the largest visible margin on TextVQA. For SmolVLM-256M the same pattern holds and the ChartQA group shows the most striking absolute lift: the original student sits near 45, the best baseline reaches roughly 68 and GRACE extends further to 72.

Answer to RQ1

GRACE improves over direct fine-tuning, same-family logit-based KD, ULD and the compared heterogeneous multi-teacher baselines in Table 5.1. The score order is same-family logits < cross-family ULD < most multi-teacher settings < GRACE. Within the multi-teacher group, GRACE scores above Uniform, SHM-CKA and REINFORCE for both student sizes across the three benchmarks.

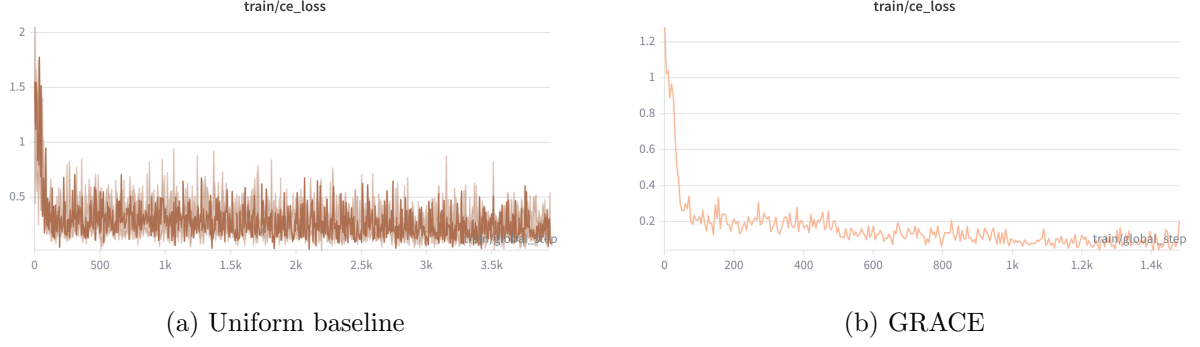
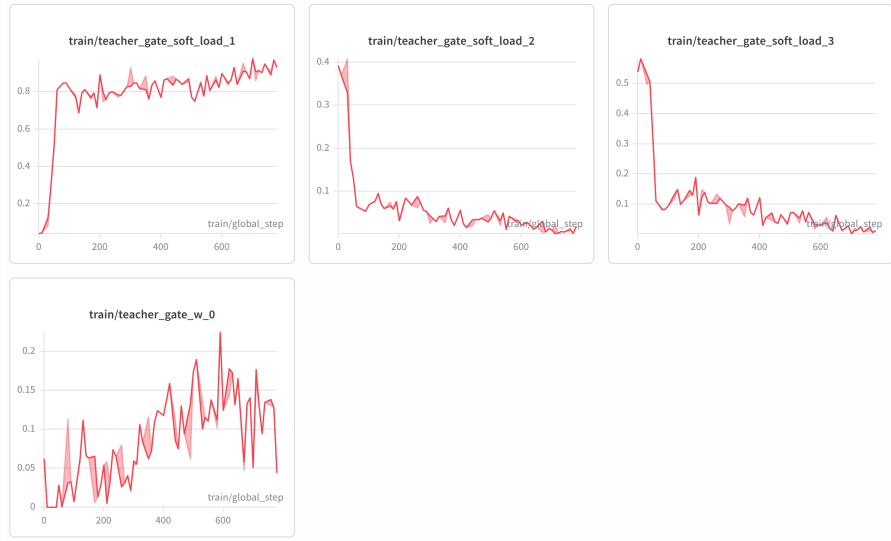
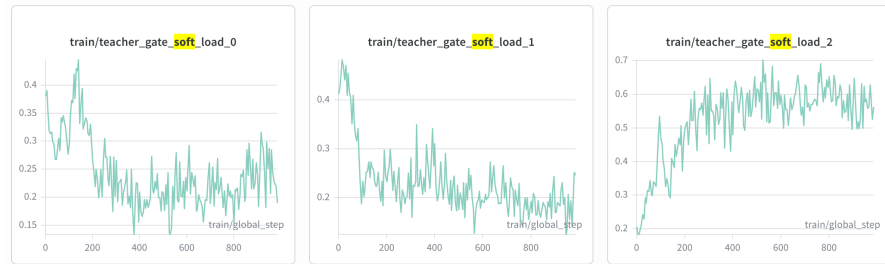


Figure 5.2: Uniform and GRACE training curves.

Figure 5.2 shows that the GRACE run is noticeably smoother and more stable than the Uniform baseline on DocVQA. Under Uniform training, the CE loss spikes repeatedly to values above 0.8 even after 2,000 steps, indicating that conflicting teacher gradients continue to disrupt updates late into training. GRACE descends more cleanly and converges to a lower final loss, consistent with the gradient-agreement filter removing teacher signals that oppose the supervised CE objective at each step.



(a) Without GRACE refinement



(b) With GRACE

Figure 5.3: Routed-teacher load examples.

Figure 5.3 shows the per-teacher soft gate load over training steps. Without the full GRACE pipeline, one teacher (load_1) rapidly captures approximately 80% of the routing weight within the first 100 steps and holds it throughout training, while the remaining two tracked teachers fall below 0.10 and stay there. With GRACE enabled, the three teacher loads remain in a 0.15 to 0.65 range throughout training, with natural variation across steps but no single teacher monopolising the pool. Table 5.2 quantifies these gains. Relative gain is computed as $(\text{GRACE} - \text{baseline})/\text{baseline}$.

Table 5.2: Relative gains of GRACE.

Benchmark	vs Original	vs Fine-tuning	vs Uniform	vs Highest non-GRACE
<i>SmolVLM-500M</i>				
TextVQA	+12.63 (+22.26%)	+12.42 (+21.80%)	+5.50 (+8.61%)	+5.38 (+8.41%)
DocVQA	+13.59 (+22.83%)	+11.15 (+18.00%)	+3.52 (+5.06%)	+3.52 (+5.06%)
ChartQA	+17.40 (+27.71%)	+17.20 (+27.30%)	+4.60 (+6.08%)	+3.32 (+4.32%)
<i>SmolVLM-256M</i>				
TextVQA	+13.56 (+28.75%)	+10.90 (+21.88%)	+6.16 (+11.29%)	+4.42 (+7.85%)
DocVQA	+11.84 (+24.34%)	+18.45 (+43.90%)	+4.78 (+8.58%)	+1.95 (+3.33%)
ChartQA	+27.28 (+60.57%)	+22.64 (+45.57%)	+8.64 (+13.57%)	+3.58 (+5.21%)

Table 5.2 shows gains over the original SmolVLM-500M student exceeding 22% on TextVQA and DocVQA and 27% on ChartQA, with similar margins over direct fine-tuning. The gain over static uniform averaging is smaller but consistent across both student sizes, with SmolVLM-256M showing the same relative pattern: +28.75%, +24.34% and +60.57%.

The teacher-gap reference is the highest teacher score in Table 4.3:

$$g_{\text{orig}} = y_T - y_{\text{orig}}, \quad g_{\text{grace}} = y_T - y_{\text{grace}}.$$

The fraction of the gap closed is $1 - g_{\text{grace}}/g_{\text{orig}}$.

Table 5.3: Gap-to-teacher analysis for SmolVLM-500M.

Benchmark	Highest teacher	Original gap	GRACE gap	Gap closed
TextVQA	79.70	22.95	10.32	55.03%
DocVQA	93.90	34.38	20.79	39.53%
ChartQA	94.24	31.44	14.04	55.34%

GRACE reduces the gap to the strongest teacher. For SmolVLM-500M, GRACE closes 55.03%, 39.53% and 55.34% of the original teacher-student gap on TextVQA, DocVQA and ChartQA. Using the strongest available teacher as the upper bound makes this gap estimate conservative.

The distilled student can outperform individual teachers. On TextVQA, GRACE reaches 69.38 and exceeds Gemma-3-4B-it at 57.80 by +11.58 points (+20.03%). On ChartQA, it exceeds InternVL3-1B at 75.30 by +4.90 points (+6.51%) and Gemma-3-4B-it at 63.60 by +16.60 points (+26.10%).

5.2 Ablation Study

The ablation builds up the full GRACE pipeline one component at a time on DocVQA, keeping all other settings fixed (variants defined in Section 4.4). The sequence is designed to separate three distinct questions: how much does teacher diversity help over a single teacher, what does sparse routing add on top of diversity and what does gradient-agreement refinement add on top of routing. Two further variants then isolate the contribution of EMA smoothing and score-based blending within the agreement step.

Table 5.4: DocVQA component ablation.

DocVQA ablation variant	SmolVLM	
	256M	500M
Cross-family ULD (single teacher)	52.24	65.28
Uniform averaging + trie-Wasserstein	55.70	69.59
Routing only + trie-Wasserstein	57.45	63.23
Full GRACE + trie-Wasserstein	60.48	73.11
w/o EMA ($\beta_{\text{EMA}} = 0$)	57.45	71.25
w/ fixed blend ($\lambda_{\text{blend}} = 1.0$)	57.97	70.45

Teacher diversity. Moving from single-teacher ULD to Uniform averaging over the full heterogeneous pool improves DocVQA by +3.46 points for SmolVLM-256M and +4.31 points for SmolVLM-500M. The gain comes purely from exposing the student to a wider range of teacher signals, with no adaptive weighting.

Routing without gradient agreement. Adding sparse top- k routing on top of Uniform produces inconsistent results. SmolVLM-256M improves by +1.75 points, but SmolVLM-500M *drops* by 6.36 points, falling below even the no-routing Uniform baseline. This is the most informative result in the ablation: a router that selects teachers by relevance alone, without checking whether their KD gradients align with the supervised CE objective, **can actively hurt training**. The larger student is apparently more sensitive to this misalign-

ment, since its capacity allows it to follow a bad routing signal further before self-correcting.

Gradient-agreement refinement. Adding gradient-agreement refinement recovers and surpasses both previous variants. Full GRACE improves over Routing only by +3.03 points for SmolVLM-256M and +9.88 points for SmolVLM-500M. The recovery for the 500M student is particularly large, which is consistent with the earlier finding that the 500M student was most harmed by unfiltered routing.

EMA smoothing and score blending. Removing EMA smoothing ($\beta_{\text{EMA}} = 0$) lowers DocVQA from 73.11 to 71.25 for SmolVLM-500M and from 60.48 to 57.45 for SmolVLM-256M, showing that per-batch agreement scores are too noisy to use directly without temporal smoothing. Setting $\lambda_{\text{blend}} = 1.0$, which removes the agreement-score softmax from the high-confidence case while keeping the threshold and fallback, also decreases both scores, confirming that score-weighted blending contributes independently of the other mechanisms.

Answer to RQ2

Teacher diversity alone accounts for a consistent gain of roughly 3 to 4 points over single-teacher ULD. Sparse routing without gradient-agreement refinement is unreliable: it helps the smaller student but hurts the larger one, demonstrating that *relevance-based selection* is insufficient on its own. Gradient-agreement refinement is the **largest single contributor**, recovering and extending the gains lost to unfiltered routing. EMA smoothing and score-based blending each provide additional, independently verifiable improvements.

5.3 Efficiency Analysis

The main online costs come from trie-Wasserstein support construction, teacher routing and gradient-agreement scoring. Trie-Wasserstein matching builds sparse token supports and accumulates trie-edge masses at answer positions. The router is an MLP over pooled student states and its cost scales with the hidden size and number of teachers. Gradient-agreement scoring adds per-teacher gradient comparisons, limited to the top- k routed teachers.

Teacher inference is **offline**: logits are cached once before training, so each update only loads them from disk. The main scaling factors are the number of teachers, the retained trie support size K_{TW} and the number of routed teachers k_{route} . An average training run on the reported configurations costs approximately **14.41 USD**, excluding one-time offline teacher logit caching and remote storage. Running the four teachers forward on the full training set to generate the cached logits is a one-time cost paid before training begins.

The 14.41 USD figure covers only the student training itself, including all routing and agreement-scoring overhead. This is low enough to be run on a single rented GPU in a few hours, placing heterogeneous multi-teacher distillation within reach of researchers without dedicated compute infrastructure.

Answer to RQ3

GRACE adds overhead from trie-Wasserstein matching, routing and gradient-agreement scoring, while offline caching replaces repeated teacher forward passes with logit loading from cache. Per answer position, trie-Wasserstein matching scales with K_{TW} , routing scales with the number of teachers N and top- k routing keeps the active teacher count bounded by k_{route} . The dominant cost is the per-step trie matching and agreement scoring, while teacher inference is paid once at caching.

5.4 Discussion

Heterogeneous distillation requires more than adding teachers to the loss: the gains come from combining cross-tokenizer output matching, sample-dependent routing and gradient reweighting after routing [15, 20, 31, 37]. Figure 5.2 shows fewer oscillations for GRACE than Uniform on DocVQA, as expected when fewer teachers receive weight per step.

Behavior of same-family logit-based KD. Same-family logit-based KD is sensitive to the divergence choice. Forward KL gives small but consistent gains, while Reverse KL performs substantially worse in this setting. Reverse KL weights its gradient by the current student distribution, so tokens the teacher scores highly but the student neglects receive weaker correction. The effect is worst on OCR-heavy and numerically sensitive tasks, where the correct answer token is often low-probability under the student’s current distribution. Reverse KL is therefore poorly suited to this setting, where the student must recover teacher probability mass on tokens it currently neglects. The claim is specific to short answer tokens and Reverse KL can behave differently in other distillation settings [59].

Cross-family ULD. This method consistently outperforms all same-family KD variants despite using only one teacher. **Sorted-rank matching** avoids the shared-vocabulary assumption entirely: comparing sorted probability vectors penalizes distributional shape differences without requiring that corresponding indices refer to the same token. ULD therefore sets the cross-family baseline that Uniform and GRACE must exceed. However, rank-based comparison discards the semantic relationship between tokens: two tokens with adjacent ranks but unrelated decoded strings receive the same low transport cost as two adjacent tokens whose strings are nearly identical. Byte-Trie Wasserstein replaces rank proximity with

byte-path proximity, giving lower transport cost to tokens with similar decoded content. This is why GRACE improves over ULD even with the same Qwen2-VL teacher in the pool.

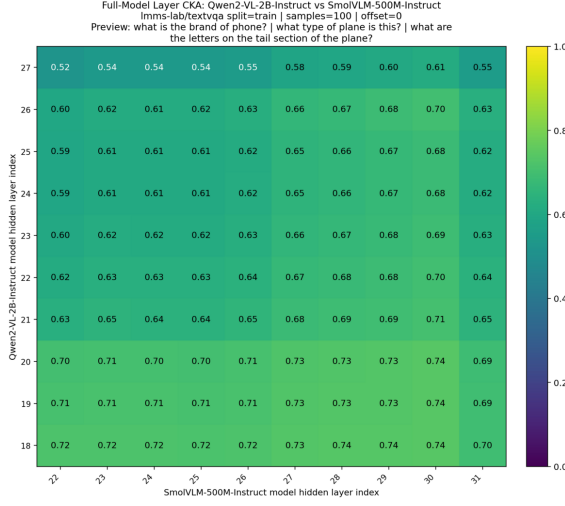
ULD measures the L1 gap between two rank-sorted probability vectors: it detects whether teacher and student distribute probability mass similarly across rank positions, but rank- k in one vocabulary need not decode to a similar string as rank- k in the other. Two token pairs with the same rank gap receive identical ULD loss regardless of whether their decoded strings are semantically related or completely unrelated. Unlike ULD, the Byte-Trie Wasserstein loss routes probability mass through the shared trie, so tokens with a longer shared byte prefix incur lower transport cost. For OCR and document tasks where teacher and student split answer strings at different subword boundaries, ULD cannot credit partial byte-level agreement, whereas Byte-Trie Wasserstein does. Table 5.5 illustrates this across four token pairs spanning the full range from exact match to no shared prefix.

Table 5.5: ULD vs. Byte-Trie Wasserstein under equal ranked masses.

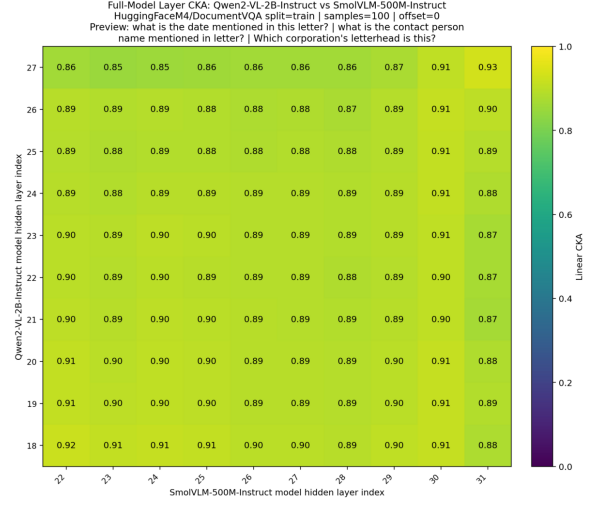
Teacher / student	Shared prefix	ℓ_{ULD}	ℓ_{TW}
<code>doc / doc</code>	<code>doc</code> (3 bytes, exact)	0.20	0.00
<code>invoice / invoices</code>	<code>invoice</code> (7 bytes)	0.20	0.05
<code>doc / document</code>	<code>doc</code> (3 bytes)	0.20	0.60
<code>abc / def</code>	none (0 bytes)	0.20	3.29

All four rows carry the same rank-space probability difference, so ULD assigns an identical loss of 0.20 in every case and cannot distinguish any of them. Byte-Trie Wasserstein, by contrast, reflects the degree of byte-path overlap: an exact decoded match (`doc/doc`) has zero cost, a near-morphological variant (`invoice/invoices`) sharing a 7-byte prefix pays only approximately 0.05, a shorter prefix overlap (`doc/document`, 3 shared bytes) costs 0.60 and a completely unrelated pair (`abc/def`, no shared bytes) pays the full 3.29. The progression tracks intuitive semantic distance in a way that rank position cannot, which is why GRACE improves over ULD even when both distill from the same teacher.

Representation matching and SHM-CKA. SHM-CKA tests whether hidden-state alignment alone is sufficient for cross-family distillation. It is competitive on ChartQA for SmolVLM-500M but trails GRACE overall [29, 30].

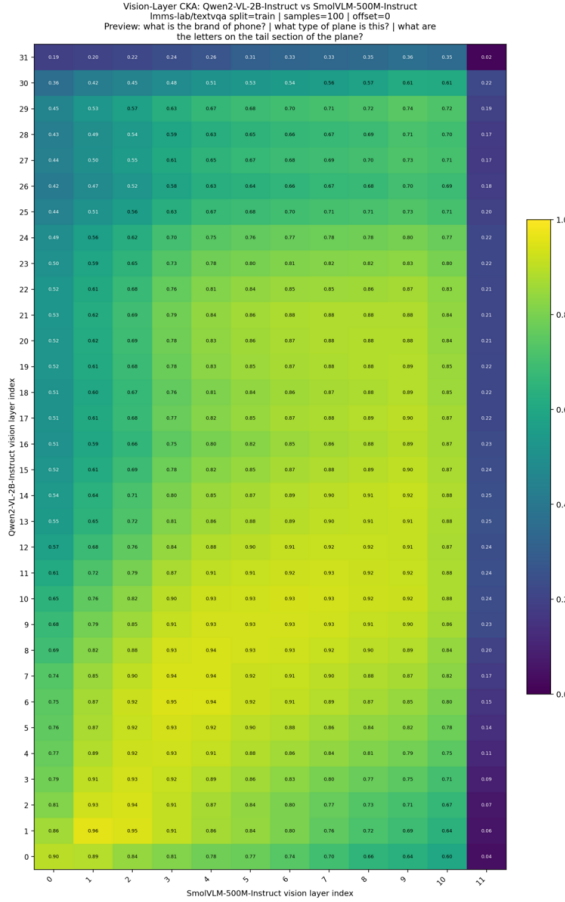


(a) TextVQA

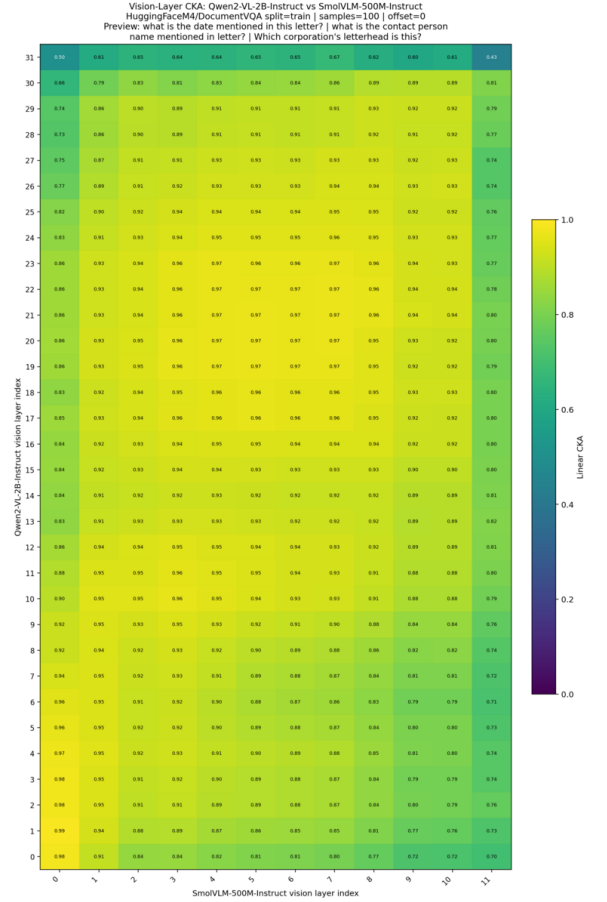


(b) DocVQA

Figure 5.4: Final-layer CKA heatmaps.



(a) TextVQA



(b) DocVQA

Figure 5.5: Vision-layer CKA heatmaps.

Figure 5.4 shows full-model layer CKA between Qwen2-VL-2B and SmolVLM-500M across the last ten language layers of each model. On TextVQA, values range from 0.52 to 0.74,

with the highest alignment at lower Qwen layers paired with later SmolVLM layers. On DocVQA, the entire matrix is compressed into a narrow 0.85 to 0.96 band with almost no variation across layer pairs, indicating that the two models are representationally similar at the language level for document inputs regardless of which layers are compared. Figure 5.5 shows the corresponding vision-layer CKA across all 32 Qwen vision layers and 13 SmolVLM vision layers. Both datasets show a clear block structure: CKA is near zero for early Qwen vision layers 0 to 10 paired with any SmolVLM layer and rises to 0.7 to 1.0 in the upper-right block where later Qwen vision layers 15 to 31 align with later SmolVLM vision layers 8 to 12. Nevertheless, SHM-CKA trails output-level distillation in Table 5.1. **High CKA does not imply strong answer prediction:** representation alignment and output calibration are largely orthogonal in heterogeneous VLM distillation.

First-order interpretation of gradient-agreement refinement. A teacher whose KD gradient conflicts with the CE gradient partially offsets the supervised update (first-order analysis in Section 3.4). Suppressing such teachers explains why gradient-agreement refinement outperforms routing alone, where routing acts as candidate selection and gradient agreement provides corrective reweighting [42].

Visual grounding.

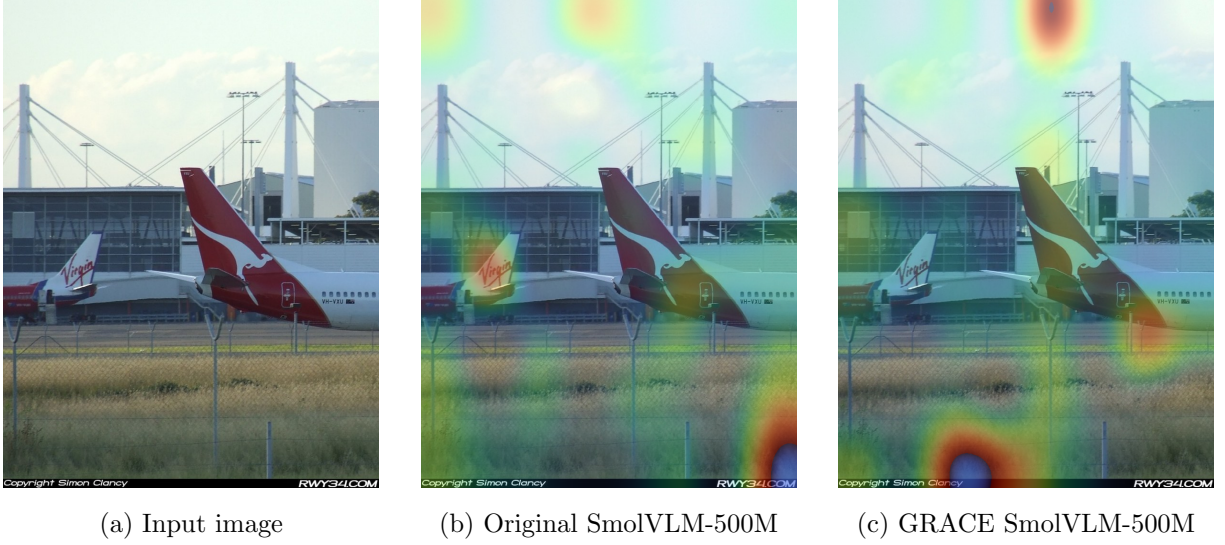


Figure 5.6: Representative TextVQA attention example.

DEX-AR attribution maps [60] aggregate token-level visual attributions into a sequence-level heatmap for autoregressive VLMs, letting us check whether distillation shifts the student’s attended regions. Figure 5.6 shows a representative TextVQA example [4] where the question asks “*who is this copyrighted by?*” and the relevant evidence is the copyright text on the lower-left border. The original SmolVLM-500M concentrates on the *Virgin* logo, while GRACE redirects attention to the copyright text, correctly finding the evidence region. Output-distribution supervision drives this shift: pushing the student toward teacher probability mass on correct answer tokens steers attention toward the image regions that support those tokens.

Conclusion and Future Work

Summary. Gradient-Routed Alignment with Capacity-aware Experts (GRACE) addresses heterogeneous VLM distillation by combining Byte-Trie Wasserstein output matching, sparse top- k teacher routing and gradient-agreement refinement [20, 31, 37]. Together these three components resolve tokenizer mismatch, adapt teacher selection per sample and filter KD gradients that conflict with the supervised CE objective. On TextVQA, DocVQA and ChartQA, GRACE outperforms same-family, single-teacher and heterogeneous multi-teacher baselines, reaching 69.38, 73.11 and 80.20 with SmolVLM-500M and 60.73, 60.48 and 72.32 with SmolVLM-256M. GRACE closes about 55% of the teacher-student gap on TextVQA and ChartQA and about 40% on DocVQA, suggesting that supervision design is a greater bottleneck than model capacity at this scale.

Limitations. Three limitations constrain the scope of the current results. **Cost-limited evaluation.** The single-run design precludes uncertainty estimates for small gaps between high-scoring baselines. An average run costs approximately 14.41 USD (Section 5.3), excluding caching cost.

Scope of validation. The study covers VQA benchmarks and SmolVLM-family students, while broader task families, other student architectures and larger teacher pools are untested. Tasks with long-form answers, multi-image inputs, video reasoning or open-ended dialogue need different forms of teacher selection and answer supervision.

Reasoning-oriented supervision. GRACE uses teacher output distributions at answer tokens as its distillation signal, filtered by gradient agreement with the supervised loss. This fits the VQA setting, where targets are short final answers, but does not transfer intermediate reasoning traces or accommodate models that generate extended reasoning at test time. Handling this requires routing teachers by intermediate reasoning quality and distilling from multi-step traces, not just final-answer distributions.

Future Work. The most immediate need is statistical: repeated runs with multiple seeds would establish which score differences between multi-teacher baselines are reliable. On the systems side, more compact logit storage and lower-cost agreement scoring would make scaling to larger teacher pools practical. Extending GRACE beyond output-level supervision to visual grounding via token alignment or cross-attention transfer and to reasoning-oriented models where intermediate traces must be distilled alongside final answers are the two substantive open directions [21, 22, 43].

References

- [1] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman and Karen Simonyan. Flamingo: a visual language model for few-shot learning, 2022.
- [2] Junnan Li, Dongxu Li, Silvio Savarese and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models, 2023.
- [3] Haotian Liu, Chunyuan Li, Qingyang Wu and Yong Jae Lee. Visual instruction tuning, 2023.
- [4] Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh and Marcus Rohrbach. Towards vqa models that can read, 2019.
- [5] Minesh Mathew, Dimosthenis Karatzas and C. V. Jawahar. Docvqa: A dataset for vqa on document images, 2021.
- [6] Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty and Enamul Hoque. Chartqa: A benchmark for question answering about charts with visual and logical reasoning, 2022.
- [7] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2.5-vl technical report, 2025.
- [8] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou and Junyang Lin. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution, 2024.
- [9] Zhe Chen, Weiyun Wang, Yue Cao, Yangzhou Liu, Zhangwei Gao, Erfei Cui, Jinguo Zhu, Shenglong Ye, Hao Tian, Zhaoyang Liu, Lixin Gu, Xuehui Wang, Qingyun Li, Yiming Ren, Zixuan Chen, Jiapeng Luo, Jiahao Wang, Tan Jiang, Bo Wang, Conghui He, Botian Shi, Xingcheng Zhang, Han Lv, Yi Wang, Wenqi Shao, Pei Chu, Zhongying

- Tu, Tong He, Zhiyong Wu, Huipeng Deng, Jiaye Ge, Kai Chen, Kaipeng Zhang, Limin Wang, Min Dou, Lewei Lu, Xizhou Zhu, Tong Lu, Dahua Lin, Yu Qiao, Jifeng Dai and Wenhai Wang. Expanding performance boundaries of open-source multimodal models with model, data and test-time scaling, 2024.
- [10] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu and Dario Amodei. Scaling laws for neural language models, 2020.
- [11] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals and Laurent Sifre. Training compute-optimal large language models, 2022.
- [12] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The llama 3 herd of models, 2024.
- [13] Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, Wenlong Huang, Yevgen Chebotar, Pierre Sermanet, Daniel Duckworth, Sergey Levine, Vincent Vanhoucke, Karol Hausman, Marc Toussaint, Klaus Greff, Andy Zeng, Igor Mordatch and Pete Florence. Palm-e: An embodied multimodal language model, 2023.
- [14] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, Pete Florence, Chuyuan Fu, Montse Gonzalez Arenas, Keerthana Gopalakrishnan, Kehang Han, Karol Hausman, Alexander Herzog, Jasmine Hsu, Brian Ichter, Alex Irpan, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Lisa Lee, Tsang-Wei Edward Lee, Sergey Levine, Yao Lu, Henryk Michalewski, Igor Mordatch, Karl Pertsch, Kanishka Rao, Krista Reymann, Michael Ryoo, Grecia Salazar, Pannag Sanketi, Pierre Sermanet, Jaspiar Singh, Anikait Singh, Radu Soricut, Huong Tran, Vincent Vanhoucke, Quan Vuong, Ayzaan Wahid, Stefan Welker, Paul Wohlhart, Jialin Wu, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu and Brianna Zitkovich. Rt-2: Vision-language-action models transfer web knowledge to robotic control, 2023.
- [15] Geoffrey Hinton, Oriol Vinyals and Jeff Dean. Distilling the knowledge in a neural network, 2015.

- [16] Xiaohan Xu, Ming Li, Chongyang Tao, Tao Shen, Reynold Cheng, Jinyang Li, Can Xu, Dacheng Tao and Tianyi Zhou. A survey on knowledge distillation of large language models, 2024.
- [17] Xiangxiang Chu, Limeng Qiao, Xinyang Lin, Shuang Xu, Yang Yang, Yiming Hu, Fei Wei, Xinyu Zhang, Bo Zhang, Xiaolin Wei and Chunhua Shen. Mobilevlm: A fast, strong and open vision language assistant for mobile devices, 2023.
- [18] Baichuan Zhou, Ying Hu, Xi Weng, Junlong Jia, Jie Luo, Xien Liu, Ji Wu and Lei Huang. Tinyllava: A framework of small-scale large multimodal models, 2024.
- [19] Andrés Marafioti, Orr Zohar, Miquel Farré, Merve Noyan, Elie Bakouch, Pedro Cuenca, Cyril Zakka, Loubna Ben Allal, Anton Lozhkov, Nouamane Tazi, Vaibhav Srivastav, Joshua Lochner, Hugo Larcher, Mathieu Morlon, Lewis Tunstall, Leandro von Werra and Thomas Wolf. Smolvlm: Redefining small and efficient multimodal models, 2025.
- [20] Nicolas Boizard, Kevin El Haddad, Pierre Colombo and Chloé Clavel. Towards cross-tokenizer distillation: The universal logit distillation loss for llms, 2024.
- [21] Fei Yuan, Linjun Shou, Jian Pei, Wutao Lin, Ming Gong, Yan Fu and Daxin Jiang. Reinforced multi-teacher selection for knowledge distillation, 2020.
- [22] Yu-Chu Yu, Chi-Pin Huang, Jr-Jen Chen, Kai-Po Chang, Yung-Hsuan Lai, Fu-En Yang and Yu-Chiang Frank Wang. Select and distill: Selective dual-teacher knowledge transfer for continual learning on vision-language models, 2024.
- [23] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger and Ilya Sutskever. Learning transferable visual models from natural language supervision, 2021.
- [24] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov and Lucas Beyer. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, page 11975–11986, 2023.
- [25] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, page 5998–6008, 2017.
- [26] Jianhua Lin. Divergence measures based on the shannon entropy. *IEEE Transactions on Information Theory*, 37(1), 1991.

- [27] Zhiyuan Fang, Jianfeng Wang, Xiaowei Hu, Lijuan Wang, Yezhou Yang and Zicheng Liu. Compressing visual-linguistic model via knowledge distillation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, page 1428–1438, 2021.
- [28] Sayantan Dasgupta and Trevor Cohn. Improving language model distillation through hidden state matching. In *International Conference on Learning Representations*, 2025.
- [29] Simon Kornblith, Mohammad Norouzi, Honglak Lee and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, page 3519–3529. PMLR, 2019.
- [30] Max Klabunde, Tobias Schumacher, Markus Strohmaier and Florian Lemmerich. Similarity of neural network models: A survey of functional and representational measures, 2023.
- [31] Gabriel Peyré and Marco Cuturi. Computational optimal transport, 2019.
- [32] Rémi Flamary and Nicolas Courty. Optimal transport for machine learning: Introduction to optimal transport, 2019. Tutorial at ISBI 2019.
- [33] Yuki Takezawa, Ryoma Sato and Makoto Yamada. Supervised tree-wasserstein distance. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, page 10086–10095. PMLR, 2021.
- [34] Tuan Le, Makoto Yamada, Kenji Fukumizu and Marco Cuturi. Tree-sliced variants of wasserstein distances. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [35] Wikimedia Commons contributors. File: Optimal transport matrix.png. Wikimedia Commons, 2018. CC BY-SA 4.0. Accessed: 2026-03-31.
- [36] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, 2017.
- [37] William Fedus, Barret Zoph and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, 2022.

- [38] DeepSeek-AI. Deepseek-v2: A strong, economical and efficient mixture-of-experts language model, 2024.
- [39] Dmitry Lepikhin, Hyounghoon Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer and Zhifeng Chen. Gshard: Scaling giant models with conditional computation and automatic sharding, 2020.
- [40] Yanqi Zhou, Tao Lei, Hanxiao Liu, Nan Du, Yanping Huang, Vincent Zhao, Andrew Dai, Zhifeng Chen, Quoc Le and James Laudon. Mixture-of-experts with expert choice routing, 2022.
- [41] Yichen Zhu and Yi Wang. Student customized knowledge distillation: Bridging the gap between student and teacher. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, page 5057–5066, 2021.
- [42] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 33, page 5824–5836, 2020.
- [43] Hailin Zhang, Defang Chen and Can Wang. Adaptive multi-teacher knowledge distillation with meta-learning, 2023.
- [44] Haodong Duan, Xinyu Fang, Junming Yang, et al. Vlmevalkit: An open-source toolkit for evaluating large multi-modality models, 2024.
- [45] Samyam Rajbhandari, Jeff Rasley, Olatunji Ruwase and Yuxiong He. Zero: Memory optimizations toward training trillion parameter models, 2020.
- [46] Tianqi Chen, Bing Xu, Chiyuan Zhang and Carlos Guestrin. Training deep nets with sublinear memory cost, 2016.
- [47] Wikimedia Commons contributors. File: Stop sign.jpg. Wikimedia Commons, 2006. Public domain. Accessed: 2026-03-06.
- [48] Wikimedia Commons contributors. File: Receiptswiss.jpg. Wikimedia Commons, 2015. CC BY-SA 3.0. Accessed: 2026-03-06.
- [49] Wikimedia Commons contributors. File: Bar chart.svg. Wikimedia Commons, 2016. CC BY-SA 4.0. Accessed: 2026-03-06.
- [50] Hugging Face TB. Smolvlm2-2.2b-instruct model card. Hugging Face, 2025. Accessed: 2026-04-02.

- [51] Qwen Team. Qwen2-vl-2b-instruct model card. Hugging Face, 2024. Accessed: 2026-04-01.
- [52] Qwen Team. Qwen2.5-vl-3b-instruct model card. Hugging Face, 2025. Accessed: 2026-04-01.
- [53] OpenGVLab. Internvl 3.0. Project blog, 2025. Accessed: 2026-04-01.
- [54] OpenGVLab. Internvl3-1b model card. Hugging Face, 2025. Accessed: 2026-05-18.
- [55] Gemma Team and Google DeepMind. Gemma 3 technical report. Technical report, 2025. Accessed: 2026-04-01.
- [56] Yifan Cui, Wei Wang, Yifei Wang, Yao Lu, Haoyu Wang, Fuzhen Zhuang, Qing He, and Chuan Shi. Sinkhorn distance minimization for knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, page 11046–11060. Association for Computational Linguistics, 2024.
- [57] Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3):229–256, 1992.
- [58] Yuqiao Wen, Zichao Li, Wenyu Du and Lili Mou. f-divergence minimization for sequence-level knowledge distillation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 10817–10834. Association for Computational Linguistics, 2023.
- [59] Taiqiang Wu, Chaofan Tao, Jiahao Wang, Runming Yang, Zhe Zhao and Ngai Wong. Rethinking kullback-leibler divergence in knowledge distillation for large language models. In *Proceedings of the 31st International Conference on Computational Linguistics*, page 5737–5755, 2025.
- [60] Walid Bousselham, Angie Boggust, Hendrik Strobelt and Hilde Kuehne. Dex-ar: A dynamic explainability method for autoregressive vision-language models, 2026.
- [61] Cloudflare. R2 object storage. Cloudflare developer documentation, 2026. Accessed: 2026-04-09.
- [62] Cloudflare. Use the aws sdk for python (boto3) with r2. Cloudflare developer documentation, 2026. Accessed: 2026-04-09.
- [63] fsspec developers. Features. fsspec documentation, 2026. Accessed: 2026-04-09.
- [64] fsspec developers. fsspec.implementations.cached. fsspec documentation, 2026. Accessed: 2026-04-09.

Appendix A

Proof of the Closed-Form Trie-Wasserstein

This appendix derives the closed-form expression used by the KD loss in Chapter 3. The result is stated for a general rooted tree [31, 33].

The proof uses the following notation. The tree $T = (V, E, r)$ is a rooted tree with root r and each edge $e \in E$ has weight $w(e) > 0$. For an edge $e = (p(e), c(e))$, define the subtree below e as

$$\Gamma(e) = \{v \in V \mid e \text{ lies on the path from } r \text{ to } v\}.$$

For vertices $u, v \in V$, the tree metric is

$$d_T(u, v) = \sum_{e \in \text{path}(u, v)} w(e).$$

μ and ν are probability measures on V . In the trie-Wasserstein setting, these are the student and teacher token distributions on the leaves of the shared byte trie, with zero mass assigned to non-leaf vertices. The Kantorovich form of the W_1 distance is

$$W_{1,T}(\mu, \nu) = \min_{\pi \in \Pi(\mu, \nu)} \sum_{u, v \in V} \pi(u, v) d_T(u, v),$$

where $\Pi(\mu, \nu)$ is the set of couplings satisfying

$$\sum_{v \in V} \pi(u, v) = \mu(u), \quad \sum_{u \in V} \pi(u, v) = \nu(v), \quad \pi(u, v) \geq 0.$$

For each edge e , the subtree imbalance is

$$\Delta(e) = \mu(\Gamma(e)) - \nu(\Gamma(e)).$$

We prove that

$$W_{1,T}(\mu, \nu) = \sum_{e \in E} w(e) |\Delta(e)|.$$

We fix any feasible coupling $\pi \in \Pi(\mu, \nu)$ and any edge $e \in E$. Removing e separates the tree into two components: $\Gamma(e)$ and $V \setminus \Gamma(e)$. Any mass transported from one side of this cut to the other must cross e .

The outward and inward crossing masses are

$$q_\pi^{\text{out}}(e) = \sum_{u \in \Gamma(e)} \sum_{v \notin \Gamma(e)} \pi(u, v), \quad q_\pi^{\text{in}}(e) = \sum_{u \notin \Gamma(e)} \sum_{v \in \Gamma(e)} \pi(u, v).$$

Thus the total mass crossing e is $q_\pi(e) = q_\pi^{\text{out}}(e) + q_\pi^{\text{in}}(e)$.

The marginal constraints imply that the source mass in $\Gamma(e)$ is

$$\mu(\Gamma(e)) = \sum_{u \in \Gamma(e)} \sum_{v \in \Gamma(e)} \pi(u, v) + q_\pi^{\text{out}}(e),$$

The target mass satisfies the same identity with q_π^{in} :

$$\nu(\Gamma(e)) = \sum_{u \in \Gamma(e)} \sum_{v \in \Gamma(e)} \pi(u, v) + q_\pi^{\text{in}}(e).$$

Subtracting:

$$\Delta(e) = q_\pi^{\text{out}}(e) - q_\pi^{\text{in}}(e), \quad |\Delta(e)| \leq q_\pi^{\text{out}}(e) + q_\pi^{\text{in}}(e) = q_\pi(e).$$

Thus every feasible transport plan must move at least $|\Delta(e)|$ mass across edge e .

Since the path between any two vertices in a tree is unique,

$$d_T(u, v) = \sum_{e \in E} w(e) \mathbf{1}\{e \in \text{path}(u, v)\}.$$

The cost of π therefore decomposes as

$$\sum_{u, v \in V} \pi(u, v) d_T(u, v) = \sum_{e \in E} w(e) q_\pi(e).$$

The inequality $q_\pi(e) \geq |\Delta(e)|$ gives

$$\sum_{u, v \in V} \pi(u, v) d_T(u, v) \geq \sum_{e \in E} w(e) |\Delta(e)|.$$

Minimizing over all feasible π gives

$$W_{1,T}(\mu, \nu) \geq \sum_{e \in E} w(e) |\Delta(e)|.$$

The lower bound is tight: we construct a coupling that achieves it. Define the signed measure

$$\sigma(v) = \mu(v) - \nu(v).$$

Since μ and ν are probability measures,

$$\sum_{v \in V} \sigma(v) = 0.$$

For each edge e , the imbalance can be written as

$$\Delta(e) = \sum_{v \in \Gamma(e)} \sigma(v).$$

Thus $\Delta(e)$ is the net surplus mass in the subtree below e .

The quantity $\Delta(e)$ can be interpreted as a signed flow across edge e . If $\Delta(e) > 0$, then $\Delta(e)$ units of surplus mass move upward from $\Gamma(e)$ toward the root. If $\Delta(e) < 0$, then $|\Delta(e)|$ units of mass move downward into $\Gamma(e)$. Because the total signed mass is zero, the total surplus equals the total deficit, so this flow can satisfy all demands. More explicitly, for each non-root vertex x , the edge to its parent is $e_x = (p(x), x)$ and the flow is

$$F(e_x) = \Delta(e_x).$$

The subtree below e_x consists of x and the subtrees below the child edges of x . Therefore,

$$F(e_x) = \sigma(x) + \sum_{y \in \text{Ch}(x)} F(e_y).$$

This identity expresses flow conservation: the net mass sent through the parent edge of x equals the local surplus at x plus the surplus received from its children. At the root, the same conservation condition follows from $\sum_{v \in V} \sigma(v) = 0$.

The signed flow F moves positive mass in σ to where σ is negative. Since T is a tree, F decomposes into transport paths from surplus vertices to deficit vertices. Let π^* be the corresponding coupling. At each vertex v , common mass $\min\{\mu(v), \nu(v)\}$ stays fixed at zero cost, so $\pi^* \in \Pi(\mu, \nu)$. By construction, the amount of mass transported across each edge e under π^* is exactly $|\Delta(e)|$. Therefore,

$$\sum_{u, v \in V} \pi^*(u, v) d_T(u, v) = \sum_{e \in E} w(e) |\Delta(e)|.$$

This construction gives the upper bound

$$W_{1,T}(\mu, \nu) \leq \sum_{e \in E} w(e) |\Delta(e)|.$$

Combining the lower and upper bounds yields

$$W_{1,T}(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|.$$

Appendix B

Implementation Details

This appendix lists the optimization and routing hyperparameters used across all distillation experiments and records the engineering details needed to reproduce the training infrastructure. The hyperparameters correspond to the evaluation protocol described in Chapter 4.

Table B.1: Experiment parameters.

Setting	Value
<i>Model setup</i>	
Student model	HuggingFaceTB/SmolVLM-500M-Instruct [19]
Teacher models	Qwen/Qwen2.5-VL-3B-Instruct [7, 52], Qwen/Qwen2-VL-2B-Instruct [8, 51], OpenGVLab/InternVL3-1B [53], google/gemma-3-4b-it [55]
Teacher weighting strategy	routing (GRACE path)
Distillation loss	Trie-Wasserstein
Trie-Wasserstein rho	0.7
Trie-Wasserstein top- k	64
Validation split	validation
<i>Hardware and infrastructure</i>	
Hardware and runtime	Single-GPU H200 or Vast.ai runs
Distributed training	DeepSpeed ZeRO-2
Numerical precision	bf16 enabled, half precision (fp16) disabled, TF32 enabled
FlashAttention-2	Enabled
Training data format	LLaVA-style multimodal instruction JSON
<i>Training</i>	
Max sequence length	16384
Training epochs	2
Per-device batch size	80
Gradient accumulation steps	1
Effective batch size	80
Gradient checkpointing	Enabled
Frozen modules	None

Continued on the next page.

Table B.1 continued from the previous page.

Setting	Value
Dataloader workers	4
Logging and checkpointing	Log every 5 steps and save every 150 steps
Seed	42
<i>Optimizer</i>	
Optimizer	adamw_torch
Adam betas	(0.9, 0.98)
Adam epsilon	1×10^{-7}
Weight decay	0.0
Base learning rate	1×10^{-5}
Vision learning rate	2×10^{-6}
Connector learning rate	1×10^{-5}
Learning-rate schedule	Cosine decay with warmup ratio 0.03
<i>GRACE hyperparameters</i>	
Student/teacher temperature	$\tau_s = \tau_t = 1.0$
KD weight	$\alpha_{\text{KD}} = 0.5$
Top- k routing	$k_{\text{route}} = 2$, with top-scoring teacher masking and selected softmax-weight renormalization
Gate entropy weight	$\lambda_{\text{ent}} = 10^{-3}$
Router z -loss weight	$\lambda_z = 10^{-3}$
GRACE warmup ratio	$\rho_g = 0.05$
GRACE softmax beta	$\beta_{\text{score}} = 4.0$
GRACE blend exponent	$\lambda_{\text{blend}} = 0.9$
GRACE EMA decay	$\beta_{\text{EMA}} = 0.8$
Score-spread fallback threshold	$\epsilon_{\text{fb}} = 0.01$
Alignment threshold	$\delta = 0.15$

B.1 Baseline Implementation

This section records the additional baseline definitions used for the comparison methods in Chapter 4, including how their teacher selection, weighting and auxiliary loss terms are implemented for fair comparison.

SHM-CKA. The baseline first loads an offline CKA matrix and, for each student layer s , selects the top- k teacher layers $\mathcal{T}_s$. Their CKA scores are normalized into soft

weights $\alpha_{s,t}$. The per-teacher layer loss is

$$\mathcal{L}_{\text{layer}}^{(i)} = \frac{1}{|S|} \sum_{s \in S} \sum_{t \in \mathcal{T}_s} \alpha_{s,t} (1 - \sqrt{\text{CKA}(\mathbf{H}_s, \mathbf{H}_t)}). \quad (\text{B.1})$$

SHM-CKA also assigns each teacher a ULD logits loss. The averaged terms are

$$\bar{\mathcal{L}}_{\text{ULD}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{ULD}}^{(i)}, \quad \bar{\mathcal{L}}_{\text{layer}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{layer}}^{(i)}. \quad (\text{B.2})$$

The final SHM-CKA objective is

$$\mathcal{L}_{\text{SHM-CKA}} = \frac{1}{3} (\mathcal{L}_{CE} + \bar{\mathcal{L}}_{\text{ULD}} + \bar{\mathcal{L}}_{\text{layer}}). \quad (\text{B.3})$$

REINFORCE. The policy uses the pooled student hidden state together with per-teacher summary features. A linear policy predicts independent Bernoulli probabilities

$$\boldsymbol{\pi} = \sigma(W \text{LN}(\mathbf{z}) + b), \quad (\text{B.4})$$

then samples a teacher mask $m_i \sim \text{Bernoulli}(\pi_i)$. If no teacher is selected, the highest-probability teacher is used as fallback. The selected mask is normalized into teacher weights

$$w_i = \frac{m_i}{\sum_{j=1}^M m_j}, \quad \mathcal{L}_{KD} = \sum_{i=1}^M w_i \mathcal{L}_{KD}^{(i)}. \quad (\text{B.5})$$

The selector is trained with the policy-gradient term

$$\mathcal{L}_{\text{pol}} = -(r - b) \log p(\mathbf{m} \mid \boldsymbol{\pi}), \quad (\text{B.6})$$

where r is the reward and b is an EMA reward baseline. The total reinforced-selection objective is

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{KD} + \lambda_{\pi} \mathcal{L}_{\text{pol}}. \quad (\text{B.7})$$

B.2 Offline Teacher-Logit Preparation

This section records the offline teacher-logit generation step used before distillation training. An average full H200 student-training run takes about 7 hours and costs about 14.41 USD at 2.059 USD/hour on Vast.ai, excluding the separate cost of offline teacher-logit generation and remote storage. Teacher logits are generated *offline* before distillation. An offline caching script runs each frozen teacher on every training example, keeps only the supervised answer-token positions and writes one `.pt` file per sample under a teacher-specific directory. Each record stores the sample index, the selected teacher logits and the

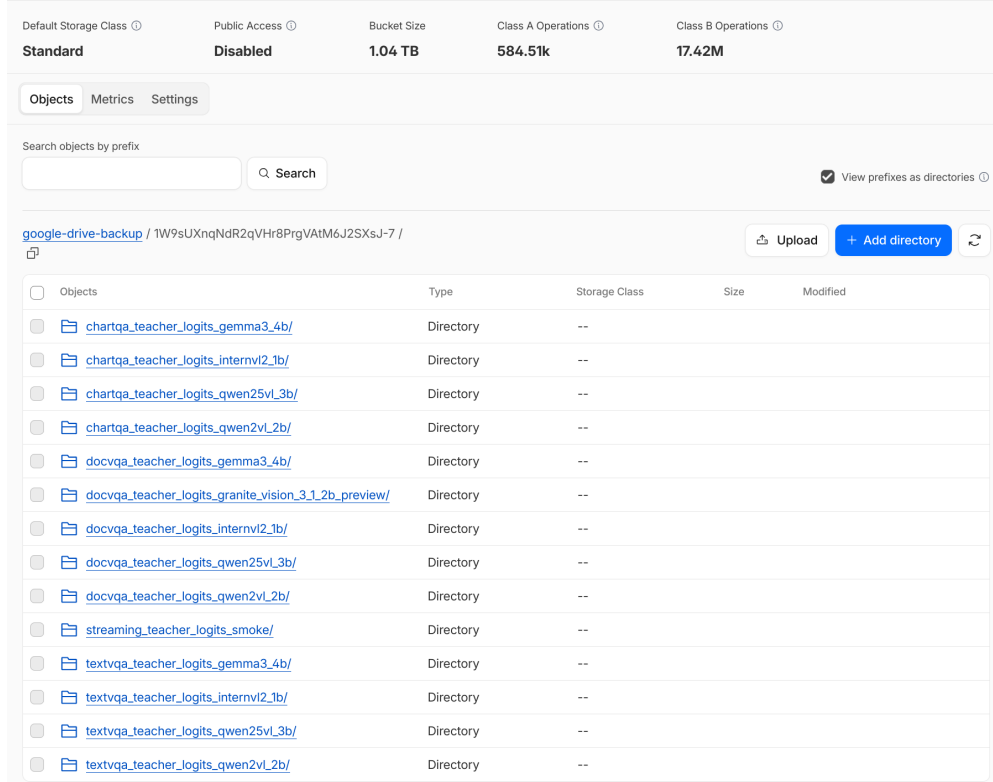


Figure B.1: R2 teacher-logit storage.

corresponding supervised labels. This keeps the online training loop simple: the student only reads precomputed teacher targets, so the teachers do not run again. After generation, the teacher-logit directories are uploaded to a **Cloudflare R2** bucket using its S3-compatible object-storage interface [61, 62]. In the active setup, the bucket already exceeds **1 TB**, so copying the full cache to an external GPU instance is not feasible. This is especially relevant on services such as Vast.ai, where local NVMe space is usually much smaller than the full teacher logit store. Figure B.1 shows this offline storage layout and highlights why the training path must stream teacher logits without staging the full cache locally.

B.3 Remote Logit Streaming During Training

The main training path does *not* download the full cache before training. Figure B.2 shows the full remote-cache path: teacher logits are generated offline and uploaded once, each training step requests a sample, the local cache serves hits directly and cache misses trigger an R2 download followed by least-recently-used eviction when the 100 GB limit is exceeded. When the launcher receives an R2 root in `distill_multi_teachers.sh`, the trainer switches to remote lazy loading. During training, `StreamingTeacherLogitsCache` resolves the teacher-specific remote object path for each requested sample, then fetches files lazily from R2 through `s3fs` and `WholeFileCacheFilesystem` [63, 64]. To keep disk usage

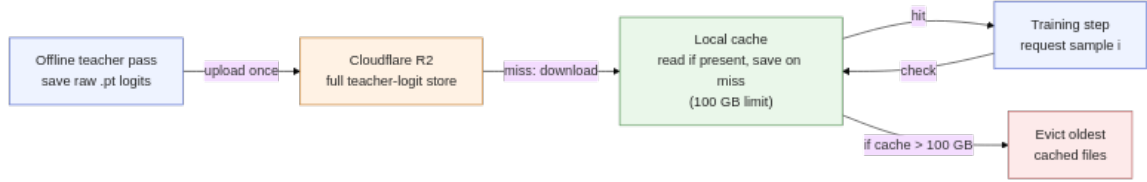


Figure B.2: Training with remote teacher logits.

bounded, the loader maintains a **100 GB** local cache. Each file is cached on first access, its access time is updated on each subsequent open and the oldest files are evicted once the cache exceeds the limit. The local cache therefore behaves as a small least-recently-used buffer over the much larger R2 store. A per-teacher file lock guards cache metadata updates so that concurrent dataloader workers do not corrupt the cache state.

Table B.2: Logit-streaming settings.

Setting	Value
Teacher-logit storage	Cloudflare R2 object storage
Teacher-logit format	Raw per-sample .pt files
Remote path template	<code>remote_root/</code> <code><dataset>_teacher_logits_</code> <code><alias>/<teacher-slug>/</code> <code><index>.pt</code>
Local cache directory	<code>/tmp/teacher-logits-</code> <code>streaming</code> for R2 reads <code>/workspace/cache</code> for local-cache reads
Local cache limit	100 GB